

A finite-volume-compatible neural WENO reconstruction framework with shared learned smoothness closure

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Abstract

1. Introduction

2. Governing Equations and Numerical Methods

2.1. Euler equations

The governing equations are the compressible Euler equations written in conservation form. In three space dimensions they are expressed as

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} + \frac{\partial \mathbf{G}}{\partial y} + \frac{\partial \mathbf{H}}{\partial z} = \mathbf{0}, \quad \mathbf{x} \in \Omega \subset \mathbb{R}^3, \quad t > 0. \quad (1)$$

The vector of conserved variables is

$$\mathbf{U} = (\rho, \rho u, \rho v, \rho w, E)^\top, \quad (2)$$

where ρ is the density, $(u, v, w)^\top$ is the velocity vector, and E is the total energy per unit volume. The inviscid flux vectors in the Cartesian coordinate directions are given by

$$\mathbf{F} = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ \rho uw \\ u(E + p) \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ \rho vw \\ v(E + p) \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} \rho w \\ \rho uw \\ \rho vw \\ \rho w^2 + p \\ w(E + p) \end{bmatrix}. \quad (3)$$

In the following, $\mathcal{F}(\mathbf{U}) = (\mathbf{F}, \mathbf{G}, \mathbf{H})$ denotes the full inviscid flux tensor. Here p denotes the pressure. The two-dimensional equations are obtained by omitting the z -momentum equation and the z -direction flux. Assuming an ideal gas, the pressure is computed from

$$p = (\gamma - 1)\rho e = (\gamma - 1)\left(E - \frac{1}{2}\rho\|\mathbf{u}\|^2\right), \quad (4)$$

where e is the specific internal energy, $\mathbf{u} = (u, v, w)^\top$, $\|\mathbf{u}\|$ is the Cartesian norm of the velocity, and $\gamma > 1$ is the ratio of specific heats. The sound speed and total enthalpy are then

$$c = \sqrt{\gamma p / \rho}, \quad H = \frac{E + p}{\rho}. \quad (5)$$

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2.2. Review of WENO schemes

We briefly review the finite-volume WENO reconstruction in the notation used below. Let v denote a scalar variable, either a conservative component or a local characteristic component of the Euler system. On a uniform one-dimensional mesh,

$$I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}], \quad \bar{v}_i = \frac{1}{\Delta x} \int_{I_i} v(x) dx. \quad (6)$$

A WENO reconstruction of formal order $2K - 1$ at a target point ξ , typically $\xi = x_{i+\frac{1}{2}}$, is built from K candidate stencils,

$$S_r(i) = \{I_{i-r}, I_{i-r+1}, \dots, I_{i-r+K-1}\}, \quad r = 0, \dots, K-1, \quad (7)$$

whose union gives the optimal large stencil

$$S^{\text{opt}}(i) = \{I_{i-K+1}, \dots, I_{i+K-1}\}. \quad (8)$$

On each $S_r(i)$, a polynomial $\mathcal{P}_r \in \mathbb{P}_{K-1}$ is defined by matching the cell averages,

$$\frac{1}{\Delta x} \int_{I_m} \mathcal{P}_r(x) dx = \bar{v}_m, \quad m = i-r, \dots, i-r+K-1. \quad (9)$$

The candidate value $\mathcal{P}_r(\xi)$ is K -th order accurate when the solution is smooth on $S_r(i)$.

The essential point of WENO is that these small-stencil reconstructions are embedded in the optimal reconstruction on the large stencil. Let $\mathcal{P}_{\text{opt}} \in \mathbb{P}_{2K-2}$ be the polynomial whose cell averages agree with $\bar{v}_m$ on all cells of $S^{\text{opt}}(i)$. The linear weights $d_r(\xi)$ are defined by

$$\mathcal{P}_{\text{opt}}(\xi) = \sum_{r=0}^{K-1} d_r(\xi) \mathcal{P}_r(\xi), \quad \sum_{r=0}^{K-1} d_r(\xi) = 1. \quad (10)$$

Thus, in smooth regions, the weighted combination of the K lower-degree candidate polynomials recovers the $(2K-1)$ -th order approximation associated with the optimal stencil.

To avoid using substencils that cross discontinuities, the linear weights are replaced by nonlinear weights. Following the Jiang–Shu construction [1], the smoothness indicator of $S_r(i)$ is

$$\beta_r = \sum_{\ell=1}^{K-1} \Delta x^{2\ell-1} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \left(\frac{d^\ell}{dx^\ell} \mathcal{P}_r(x) \right)^2 dx, \quad r = 0, \dots, K-1. \quad (11)$$

The nonlinear weights are then

$$\alpha_r(\xi) = \frac{d_r(\xi)}{(\varepsilon + \beta_r)^a}, \quad \omega_r(\xi) = \frac{\alpha_r(\xi)}{\sum_{s=0}^{K-1} \alpha_s(\xi)}, \quad r = 0, \dots, K-1, \quad (12)$$

where $\varepsilon > 0$ prevents division by zero and $a = 2$ is the classical choice. If the data are smooth on all substencils, then $\omega_r(\xi)$ approaches $d_r(\xi)$ and the optimal large-stencil accuracy is recovered. If a discontinuity lies in $S_r(i)$, the corresponding β_r becomes large and the stencil is automatically down-weighted.

The reconstructed interface traces are therefore written as

$$v_{i+\frac{1}{2}}^- = \sum_{r=0}^{K-1} \omega_r^-(x_{i+\frac{1}{2}}) \mathcal{P}_r^-(x_{i+\frac{1}{2}}), \quad v_{i+\frac{1}{2}}^+ = \sum_{r=0}^{K-1} \omega_r^+(x_{i+\frac{1}{2}}) \mathcal{P}_r^+(x_{i+\frac{1}{2}}). \quad (13)$$

Here the superscripts $-$ and $+$ denote reconstructions from the left and right sides of the interface, respectively. For systems of conservation laws, the scalar reconstruction above is applied componentwise or, more commonly for

compressible flows with shocks, in local characteristic variables. The reconstructed characteristic states are then transformed back to conservative variables and supplied to a Riemann solver.

In multiple space dimensions, a finite-volume method requires surface-averaged numerical fluxes. Hence the reconstructed states must be available at Gaussian quadrature points on each cell face, not only at the face center. Let η_α and λ_α , $\alpha = 1, \dots, N_q$, be the Gauss–Legendre nodes and weights on $[-1/2, 1/2]$, with N_q chosen consistently with the target spatial order. For an x -normal face in three dimensions, the quadrature points are

$$\mathbf{x}_{\alpha\beta}^x = (x_{i+\frac{1}{2}}, y_j + \eta_\alpha \Delta y, z_\ell + \eta_\beta \Delta z), \quad \alpha, \beta = 1, \dots, N_q. \quad (14)$$

The corresponding face-averaged flux is approximated by

$$\widehat{\mathbf{F}}_{i+\frac{1}{2},j,\ell} = \sum_{\alpha=1}^{N_q} \sum_{\beta=1}^{N_q} \lambda_\alpha \lambda_\beta \mathcal{H}_x(\mathbf{U}_{i+\frac{1}{2},j,\ell,\beta}^-, \mathbf{U}_{i+\frac{1}{2},j,\ell,\beta}^+), \quad (15)$$

where $\mathcal{H}_x$ is a consistent numerical flux in the x direction. The fluxes $\widehat{\mathbf{G}}$ and $\widehat{\mathbf{H}}$ are obtained by cyclic permutation of the coordinate directions. In two dimensions, the summation over β and the reconstruction in the z direction are omitted.

The point values in (15) are obtained by a dimension-by-dimension WENO reconstruction [2]. For the x -normal face, one first reconstructs in the normal direction and then in the tangential directions:

$$\overline{\mathbf{U}}_{i+\frac{1}{2},j,\ell}^{x,\pm} = \mathcal{R}_x(\{\overline{\mathbf{U}}_{m,j,\ell}\}_m)(x_{i+\frac{1}{2}}), \quad (16)$$

$$\overline{\mathbf{U}}_{i+\frac{1}{2},j,\ell}^{xy,\pm} = \mathcal{R}_y(\{\overline{\mathbf{U}}_{i+\frac{1}{2},m,\ell}^{x,\pm}\}_m)(y_j + \eta_\alpha \Delta y), \quad (17)$$

$$\mathbf{U}_{i+\frac{1}{2},j,\ell,\beta}^\pm = \mathcal{R}_z(\{\overline{\mathbf{U}}_{i+\frac{1}{2},j,\ell,\beta}^{xy,\pm}\}_m)(z_\ell + \eta_\beta \Delta z). \quad (18)$$

Here $\mathcal{R}_x$, $\mathcal{R}_y$, and $\mathcal{R}_z$ denote the same one-dimensional WENO operator defined above, evaluated at the appropriate target point. Consequently, the linear weights $d_r(\xi)$ and the nonlinear weights $\omega_r(\xi)$ depend on the particular Gaussian node at which the reconstruction is performed. This tensor-product procedure converts cell averages into high-order pointwise left and right states.

3. WENO-SR Reconstruction and GPU Implementation

3.1. Finite-volume WENO-SR reconstruction

We now describe the learned nonlinear weighting strategy used in this work. The objective is not to replace the finite-volume WENO reconstruction by a black-box approximation, but to learn a compact and reusable map that identifies the relative quality of the candidate substencils. The polynomial candidates, the optimal linear weights, the conservative update, and the Riemann solver remain those of the underlying finite-volume method. Only the mechanism that converts local stencil information into nonlinear WENO weights is modified.

In this subsection we denote by r the number of candidate substencils of a WENO scheme of order $2r - 1$; this corresponds to K in the notation of the previous subsection. Given a scalar stencil

$$\mathbf{q} = (q_0, q_1, \dots, q_{2r-2})^\top, \quad (19)$$

where q_j denotes either a conservative component or, in the characteristic formulation, a local characteristic component, we construct the neural-network input vector

$$\mathbf{z} = (\widehat{\delta}_0, \dots, \widehat{\delta}_{r-1}, \gamma_s, s)^\top \in \mathbb{R}^{r+2}. \quad (20)$$

The first r entries measure the relative smoothness of each candidate substencil. Let the Jiang–Shu smoothness indicator on substencil k be written as a quadratic form in a set of linear smoothness modes,

$$\beta_k^{\text{JS}} = \sum_{m=1}^{r-1} c_m D_{k,m}^2, \quad k = 0, \dots, r-1, \quad (21)$$

where $D_{k,m}$ are stencil-dependent linear combinations of the data and $c_m > 0$ are the corresponding coefficients. Instead of feeding β_k^{JS} or its square root to the network, we use an unsquared, L^1 -type badness indicator,

$$\delta_k = \sum_{m=1}^{r-1} c_m |D_{k,m}|, \quad k = 0, \dots, r-1. \quad (22)$$

This form preserves the information about which substencil is less regular while avoiding the stronger compression produced by squaring. The indicators are normalized within the full stencil,

$$\delta_{\max} = \max_{0 \leq k \leq r-1} \delta_k, \quad \widehat{\delta}_k = \frac{\delta_k}{\delta_{\max} + \varepsilon_\delta}. \quad (23)$$

Thus, the first part of the input tells the network which substencils are relatively bad or relatively smooth in the current stencil.

The next input is a shock/smoothness feature γ_s , designed to distinguish smooth high-frequency oscillations from unresolved nonsmooth structures. Let

$$\Delta q_j = q_{j+1} - q_j, \quad \Delta^m q_j = \Delta(\Delta^{m-1} q_j) \quad (24)$$

be the forward differences. For a consecutive local window

$$I_{j,\ell} = \{q_j, \dots, q_{j+\ell}\}, \quad \ell = 2, \dots, r-1, \quad (25)$$

we define the first-order total variation and the high-order nonlinear variation as

$$V_1(I_{j,\ell}) = \sum_{a=0}^{\ell-1} |\Delta q_{j+a}|, \quad V_H(I_{j,\ell}) = \sum_{m=2}^{\ell} \sum_{a=0}^{\ell-m} |\Delta^m q_{j+a}|. \quad (26)$$

The detector is then

$$\gamma_s = \min \left(1, \max_{\ell=2, \dots, r-1} \max_{j=0, \dots, 2r-2-\ell} \frac{V_H(I_{j,\ell})}{V_1(I_{j,\ell}) + \varepsilon_\gamma} \right). \quad (27)$$

For nearly linear data the high-order differences are small and $\gamma_s \approx 0$. Near shocks, kinks, or unresolved oscillations, the high-order variation becomes comparable to the first-order variation and $\gamma_s = O(1)$.

The final input is an absolute scale feature. This feature prevents normalized badness ratios from misclassifying floating-point plateau regions as nonsmooth merely because all δ_k are close to round-off. We set

$$q_{\text{scale}} = \max(|q_0|, \dots, |q_{2r-2}|, 1), \quad \eta = \max \left(\frac{\delta_{\max}}{q_{\text{scale}}}, 10^{-30} \right), \quad (28)$$

and define

$$s = \text{clip} \left(\frac{\log_{10} \eta + \kappa_s}{\kappa_s}, 0, 1 \right). \quad (29)$$

The scale range is order specific: $\kappa_s = 4$ for the WENO5-SR model and $\kappa_s = 16$ for WENO7-SR. In particular, the five WENO5 input features are unchanged throughout the distance-balanced training described below.

The neural network is a fully connected multilayer perceptron

$$\mathcal{N}_\theta : \mathbb{R}^{r+2} \longrightarrow \mathbb{R}^r, \quad \mathbf{a} = \mathcal{N}_\theta(\mathbf{z}), \quad (30)$$

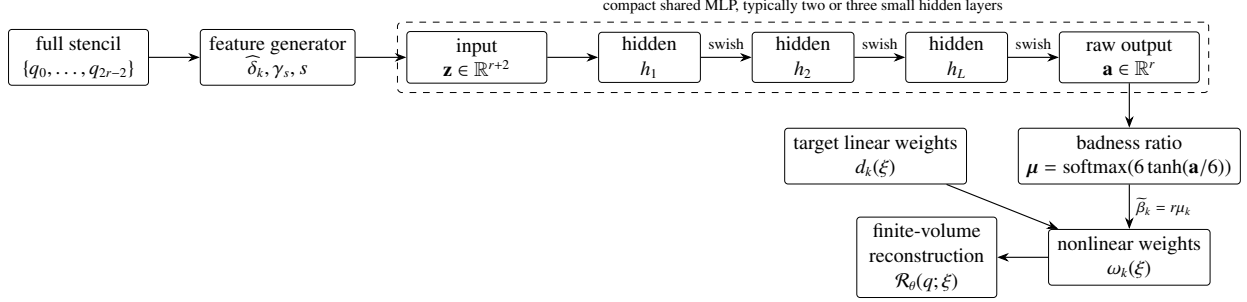


Figure 1: Compact learned-WENO weighting architecture. The network maps the generic feature vector $\mathbf{z} \in \mathbb{R}^{r+2}$ to a shared substencil badness ratio in $\mathbb{R}^r$. The final nonlinear weights are then obtained by combining this learned ratio with the target-dependent linear weights $d_k(\xi)$, so the same network can be reused for face-center and Gaussian-point reconstructions.

with the swish activation function

$$\phi(x) = x\sigma(x), \quad \sigma(x) = \frac{1}{1 + \exp(-x)} \quad (31)$$

in the hidden layers. The framework does not require a large network: in our numerical experiments, small multilayer perceptrons with two or three hidden layers and only a modest number of neurons per layer were sufficient to learn the substencil selection mechanism. This compact design reduces the cost of stencil-wise evaluations and avoids the overfitting tendency observed with larger networks. In general, the architecture can be written as

$$(r+2) \rightarrow h_1 \rightarrow \dots \rightarrow h_L \rightarrow r, \quad L = 2 \text{ or } 3, \quad (32)$$

where h_ℓ are small hidden-layer widths. The specific widths used for a particular WENO order are reported together with the numerical configuration.

The raw output of the network is not used directly as the WENO weights. Instead, it is first converted into a positive normalized badness ratio,

$$\boldsymbol{\mu} = \text{softmax}\left(6 \tanh\left(\frac{\mathbf{a}}{6}\right)\right), \quad \sum_{k=0}^{r-1} \mu_k = 1, \quad \mu_k > 0. \quad (33)$$

Reflection equivariance is imposed by construction rather than learned from data augmentation. Let P_z reverse the normalized-difference channels while leaving (γ_s, s) unchanged, and let P_μ reverse the substencil outputs. Thus, P_z exchanges $\widehat{\delta}_0$ and $\widehat{\delta}_2$ for WENO5-SR and reverses all four $\widehat{\delta}_k$ channels for WENO7-SR. The ratio used by every learned reconstruction is

$$\boldsymbol{\mu}_\theta^{\text{sym}}(\mathbf{z}) = \frac{1}{2} [\boldsymbol{\mu}_\theta(\mathbf{z}) + P_\mu \boldsymbol{\mu}_\theta(P_z \mathbf{z})]. \quad (34)$$

Both branches share the same parameters, so $\boldsymbol{\mu}_\theta^{\text{sym}}(P_z \mathbf{z}) = P_\mu \boldsymbol{\mu}_\theta^{\text{sym}}(\mathbf{z})$ up to round-off. This two-evaluation average is used during both training and inference for WENO5-SR and WENO7-SR. The bounded logits avoid excessively large ratios during inference. We then define an effective badness vector

$$\widetilde{\beta}_k = r\mu_k, \quad k = 0, \dots, r-1. \quad (35)$$

For each reconstruction target point ξ , the final nonlinear WENO weights are obtained by combining this shared badness vector with the corresponding optimal linear weights $d_k(\xi)$,

$$\alpha_k(\xi) = \frac{d_k(\xi)}{(\widetilde{\beta}_k + \varepsilon_\beta)^2}, \quad \omega_k(\xi) = \frac{\alpha_k(\xi)}{\sum_{\ell=0}^{r-1} \alpha_\ell(\xi)}. \quad (36)$$

This construction is particularly useful in finite-volume schemes. The neural network is independent of the reconstruction target and is therefore shared by face-center and Gaussian-point reconstructions, while the target-point dependence is retained through $d_k(\xi)$. Consequently, the same learned local badness estimate can be used for all quadrature points required by the multidimensional finite-volume update.

The two networks used below are deliberately compact. WENO5-SR uses the architecture $5 \rightarrow 10 \rightarrow 6 \rightarrow 6 \rightarrow 3$, whereas WENO7-SR uses $6 \rightarrow 24 \rightarrow 16 \rightarrow 16 \rightarrow 4$. The last affine layer is initialised to zero, so that $\mu = (1/r)\mathbf{1}$ and the initial model is exactly the optimal linear scheme. These networks are smaller than the $4 \rightarrow 16 \rightarrow 16 \rightarrow 16 \rightarrow 2$ WENO3-NN model of Bezgin et al. [3] and the $4 \rightarrow 32 \rightarrow 16 \rightarrow 16 \rightarrow 3$ WENO5-NN model of Nogueira et al. [4]. The comparison is not intended to equate the different input embeddings; it illustrates that the present shared badness representation does not require a large inference model.

3.2. End-to-end distance-balanced finite-volume training

Training only on isolated canonical stencils constrains a reconstruction map at selected local states, but does not constrain the distribution of states created by repeated application of the resulting finite-volume operator. This distinction is mild for a short, compact reconstruction and becomes increasingly important as the stencil width and propagation time increase. The static reconstruction datasets used in earlier neural WENO studies [3, 4] remain useful local diagnostics, but they do not by themselves control accumulated dissipation, phase error, or variation growth. We therefore train through the time-marching operator and evaluate all auxiliary reconstruction terms online on the same randomly generated analytic profiles. There is no separate offline pretraining stage or stored teacher-weight dataset in the selected model.

The selected WENO5-SR and WENO7-SR models are trained through their complete finite-volume time-marching operators, without a precomputed database of teacher weights. We use the periodic scalar problem

$$u_t + au_x = 0, \quad x \in [0, 1], \quad a \in \{-1, 1\}, \quad (37)$$

because its exact translated solution isolates the reconstruction and time-integration operators from Riemann-solver and equation-of-state errors. It also provides a direct gradient path from the evolved finite-volume state to the reconstruction network, without introducing a learned signal through a Riemann solver. The training state consists exclusively of cell averages,

$$\bar{u}_i(t) = \frac{1}{\Delta x} \int_{x_{i-1/2}}^{x_{i+1/2}} u_0(x - at) dx. \quad (38)$$

Continuous components are integrated by 15-point Gauss–Legendre quadrature and box components by exact geometric overlap. Centre samples are never used as finite-volume targets.

The on-the-fly profile generator contains seven families: smooth Fourier waves, Gaussian pulses, square waves, triangles, semi-ellipses, multiple jumps, and random composites of these components. Their sampling probabilities are

$$(0.25, 0.15, 0.15, 0.15, 0.10, 0.10, 0.10), \quad (39)$$

respectively. Amplitudes, positions, widths, phases, and the number of active components are randomized. Positive- and negative-velocity samples occur in equal numbers in every batch. None of the GSTE, Euler, or multidimensional test configurations used in Section 4 is included in the training generator.

Let $\Phi_\theta^{\Delta t}$ denote one complete finite-volume time step formed by the learned spatial reconstruction and the time integrator selected for a given scheme. The order-matched classical operator $\Phi_{JS}^{\Delta t}$ differs only in its nonlinear WENO weights. This formulation is not tied to a particular Runge–Kutta formula: WENO5-SR is coupled to SSPRK3 in the present experiments, WENO7-SR is coupled to a fourth-order SSP Runge–Kutta method, and the same construction can be used with a one-step ADER update. Starting from the exact cell averages, the learned trajectory is advanced autoregressively:

$$\bar{\mathbf{u}}_\theta^{m+1} = \Phi_\theta^{\Delta t}(\bar{\mathbf{u}}_\theta^m), \quad \bar{\mathbf{u}}_\theta^0 = \bar{\mathbf{u}}_{\text{ex}}(0). \quad (40)$$

The state is never detached. Supervision is applied after each complete time step, where the numerical and analytic cell averages represent the same physical time. Internal stages are not assigned artificial exact targets. Consequently, the training principle remains well defined for either multistage Runge–Kutta or one-step space–time integrators. For

sample b , the normalized state error is

$$e_{\theta,b}^m = \left[\frac{1}{N} \sum_i \left| \frac{\bar{u}_{\theta,b,i}^m - \bar{u}_{\text{ex},b,i}^m}{q_{\text{scale},b}} \right|^4 \right]^{1/2} + 0.05 \frac{1}{N} \sum_i \left(\frac{\bar{u}_{\theta,b,i}^m - \bar{u}_{\text{ex},b,i}^m}{A_b} \right)^2, \quad (41)$$

where A_b is the value range, bounded below by $10^{-6} q_{\text{scale},b}$. Errors are first averaged within each profile family present in a batch and then across families, preventing the most common family from dominating the update.

Two non-regression guards compare the learned and order-matched WENO-JS trajectories at every step. The global guard penalizes $[e_{\theta,b}^m - e_{\text{JS},b}^m]_+$. The local guard applies the same hinge to periodic eight-cell RMS error windows. For either guard, violations are aggregated as

$$\mathcal{A}(v) = 0.25 \text{ mean}_{\text{family}}(v) + 0.75 \text{ CVaR}_{0.25}(v), \quad (42)$$

so both typical and worst-quartile regressions contribute. The classical trajectory is used only as a detached non-regression baseline; it does not provide target weights or target solutions.

The trajectory objective is complemented by four terms evaluated on the same analytic profiles. A face-path term compares the upwind face reconstructed from the evolving learned state with the exact face value. An exact reconstruction term evaluates the face and Gaussian-point targets required by the corresponding finite-volume scheme on the exact cell-average state. For WENO5 these are $\xi/\Delta x \in \{-1/2, 1/2, -\sqrt{3}/6, \sqrt{3}/6\}$; WENO7 uses its order-matched candidate polynomials and optimal weights at the same target locations. A flat-region term matches second differences where the exact variation is below $2 \times 10^{-3} A_b$. Finally,

$$\mathcal{L}_{\text{TV}} = \left[\frac{\text{TV}(\bar{\mathbf{u}}_{\theta}^m) - \text{TV}(\bar{\mathbf{u}}_{\text{ex}}^m)}{A_b} \right]_+ \quad (43)$$

penalizes only excess variation. The resulting primary objective is

$$\begin{aligned} \mathcal{L}_{\text{primary}} = & \mathcal{L}_{\text{state}} + 0.04 \mathcal{L}_{\text{face}} + 0.15 \mathcal{L}_{\text{recon}} + 0.05 \mathcal{L}_{\text{flat}} + 0.03 \mathcal{L}_{\text{TV}} \\ & + \mathcal{L}_{\text{guard}}^{\text{global}} + \mathcal{L}_{\text{guard}}^{\text{local}}. \end{aligned} \quad (44)$$

An additional guard-only trajectory, limited to the first 40 steps and weighted by 0.25, preserves short-range non-regression even when the primary trajectory is very long.

Training uses $N = 96$ cells and the single CFL number $\nu = 0.5$. Propagation distances

$$D/\Delta x \in \{2, 8, 64, 128, 256, 512, 1024\} \quad (45)$$

are visited exactly once in a shuffled seven-iteration cycle. For both training instances, $\nu = 0.5$ converts these distances to $\{4, 16, 128, 256, 512, 1024, 2048\}$ complete time steps. To control memory while retaining equal distance-level sampling probability, the associated batch sizes are $\{32, 32, 16, 8, 4, 2, 2\}$. Activation checkpointing recomputes intermediate states during back-propagation, preserving the complete long-range gradient without detaching the autoregressive state.

The network starts from the exact optimal linear weights because its final affine layer is zero initialized. Adam is configured for 2×10^5 iterations with a cosine learning-rate schedule from 10^{-4} to 2×10^{-5} . Gradient norms are clipped at one, and an update is skipped if its pre-clipping norm exceeds ten. Models and online validation fields are saved every 250 iterations. The independently selected checkpoints used in Section 4 are WENO5-SR-FP64 at iteration 12250, WENO5-SR-FP32 at iteration 16500, and WENO7-SR-FP64 at iteration 16750. Thus, the reported training is end to end with respect to the evolved finite-volume state while remaining teacher-free: all supervision is generated analytically from the sampled initial condition, and no Euler trajectory or learned reference model enters the optimization.

3.3. GPU inference and mixed precision

All flow states, reconstructed conservative variables, numerical fluxes, and Runge–Kutta updates are stored in double precision. For WENO5-SR the compact MLP is evaluated directly inside NVIDIA Warp CUDA kernels, avoiding a separate framework launch for every stencil. The WENO7-SR implementation evaluates the badness ratios in batched PyTorch operations using zero-copy views of the Warp state arrays; Warp kernels subsequently perform the characteristic projection, WENO reconstruction, HLLC flux, and fourth-order Runge–Kutta update. Thus PyTorch supplies only the learned local closure, not the PDE solver.

For all Euler tests, the HLLC outer-wave estimates use the direct two-state bounds [5],

$$S_L = \min(u_{n,L} - a_L, u_{n,R} - a_R), \quad S_R = \max(u_{n,L} + a_L, u_{n,R} + a_R), \quad (46)$$

where u_n is the velocity normal to the face. A common $\varepsilon_{\text{HLLC}} = 10^{-16}$ is used when evaluating the sound speed and when a contact- or star-state denominator is numerically indistinguishable from zero. Density and pressure entering those local HLLC expressions are bounded below by the same value. This machine-scale safeguard is applied identically to every method; no method-dependent repair or fallback Riemann solver is used.

To assess reduced-precision inference on consumer GPUs, we independently train a mixed-precision WENO5-SR model from the same linear initialization with the same distance-balanced objective. Its MLP parameters and hidden activations are `float32`; the five input features are cast to single precision before the network evaluation, and the predicted badness ratios are cast back to double precision before weight normalisation and reconstruction. The state, Riemann solver, SSPRK3 stages, and diagnostics remain in double precision. This is a mixed-precision implementation rather than an integer-quantised model.

Although the implementation provides an optional ENO cutoff for diagnostic purposes, all numerical results reported in this paper use the raw learned weights without post-processing.

4. Numerical Results

This section evaluates the accuracy, robustness, and computational cost of the proposed WENO-SR reconstruction. Unless stated otherwise, Euler calculations use local characteristic reconstruction and the HLLC approximate Riemann solver. The three-dimensional shock–bubble benchmark instead uses the validated EVILIN flux throughout, because its high-resolution WENO7–ADER4 reference was computed with the same Riemann solver; this keeps the flux formulation fixed across the resolution and reconstruction comparisons. Unless stated otherwise, two-dimensional calculations use a CFL number of 0.4. No inference-time ENO cutoff is applied. Results are reported only for runs that reach the prescribed final time without non-finite values or negative density or pressure. The HLLC outer waves use Eq. (46) with $\varepsilon_{\text{HLLC}} = 10^{-16}$ in every newly computed result.

The experiments are generated by the isolated and reproducible workflow in `for_paper_results`. The spatial reconstruction is not tied to one time discretisation. In the present benchmarks, WENO5 calculations use a finite-volume SSPRK3 solver implemented with NVIDIA Warp, whereas WENO7 calculations use a finite-volume fourth-order SSP Runge–Kutta formulation. The WENO5-SR network is evaluated inside the Warp kernels. In the standard WENO7-SR path, PyTorch evaluates the badness ratios in batches, while Warp performs reconstruction, HLLC flux evaluation, and time integration. For the 1200×300 double-Mach calculation, the same FP64 network map is evaluated by a fused Warp kernel to avoid fine-grained framework-launch overhead; a direct cross-check against the PyTorch path gives a relative discrepancy of 9.3×10^{-16} in the initial semi-discrete right-hand side. Classical and learned methods of a given order otherwise share the same solver path; only the nonlinear weighting rule differs. The comparisons also include the WENO-Z weighting of Borges et al. [6] and its high-order extension and analysis by Castro et al. [7] as additional order-matched classical baselines. These schemes are especially relevant in smooth regions because their global smoothness correction drives the nonlinear weights toward the optimal linear weights more rapidly near critical points. With $r = 3$ and 4 substencils for WENO5 and WENO7, respectively, the nonlinear weights use $p_Z = r - 1$. We take $\varepsilon = h^2$ and $\tau_5 = |\beta_0 - \beta_2|$ for WENO5-Z, and $\varepsilon = h^3$ and $\tau_7 = |-\beta_0 - 3\beta_1 + 3\beta_2 + \beta_3|$ for WENO7-Z. Directional mesh spacings are used in multidimensional calculations. Common factors in the implemented smoothness indicators are retained together with the correspondingly scaled ε , leaving the weights algebraically unchanged.

Table 1: Methods used in the Euler benchmarks. The state and flow solver are double precision in every case; “FP32 MLP” refers only to the network parameters and hidden activations.

Method	Spatial weighting	Time integrator	MLP precision
WENO5-JS-RK3	Jiang–Shu	SSPRK3	–
WENO5-Z-RK3	WENO-Z	SSPRK3	–
WENO5-SR-RK3	learned shared badness	SSPRK3	FP64
WENO5-SR-FP32-RK3	learned shared badness	SSPRK3	FP32
WENO7-JS-RK4	Jiang–Shu	fourth-order SSP RK	–
WENO7-Z-RK4	WENO-Z	fourth-order SSP RK	–
WENO7-SR-RK4	learned shared badness	fourth-order SSP RK	FP64

The selected checkpoints are distance-balanced WENO5-SR-FP64 at optimisation step 12250, independently trained WENO5-SR-FP32 at step 16500, and distance-balanced WENO7-SR-FP64 at step 16750. Their file hashes and complete runtime configuration are recorded in the accompanying experiment manifest. The GSTE calculation was not sampled by the optimiser, but was used as an external long-time validation problem during checkpoint selection; the Euler configurations below remained outside the scalar-advection training distribution.

4.1. Long-time advection of the GSTE profile

We first solve $u_t + u_x = 0$ on $[-1, 1]$ with periodic boundaries. The initial condition is the standard composite Gaussian–square–triangle–semi-ellipse (GSTE) profile of Bezgin et al. [3]. Cell averages are initialised by Gaussian quadrature on 200 control volumes and evolved to $t = 10$, corresponding to five traversals of the domain. All seven methods use CFL 0.6; the WENO5 variants use SSPRK3 and the WENO7 variants use the fourth-order SSP Runge–Kutta update.

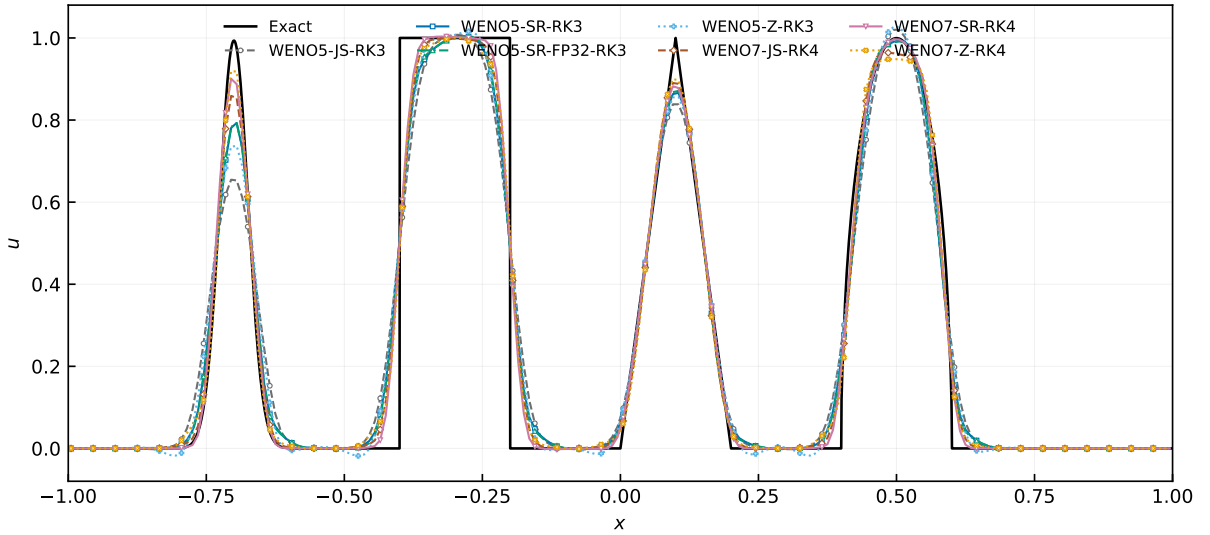


Figure 2: Long-time advection of the GSTE profile on 200 control volumes at $t = 10$. The selected learned models and the order-matched JS and Z schemes are plotted against the exact periodic translation.

Table 2: Scalar L_1 error for the GSTE advection problem at $N = 200$ and $t = 10$. Errors are evaluated against the exact periodic finite-volume translation.

Method	L_1 error
WENO5-JS-RK3	4.977×10^{-2}
WENO5-Z-RK3	4.110×10^{-2}
WENO5-SR-RK3	3.756×10^{-2}
WENO5-SR-FP32-RK3	3.645×10^{-2}
WENO7-JS-RK4	2.974×10^{-2}
WENO7-Z-RK4	3.678×10^{-2}
WENO7-SR-RK4	2.342×10^{-2}

Despite the fact that the composite GSTE profile is excluded from training, all three learned models remain stable over all 1667 time steps. WENO-Z is already beneficial in this long-time transport test: WENO5-Z reduces the WENO5-JS error by 17.4%, consistently with the lower-dissipation behaviour reported for the original WENO-Z construction [6, 7]. Each learned model nevertheless gives a smaller L_1 error than both order-matched classical baselines in Table 2. WENO5-SR-FP32 is the most accurate WENO5 variant, while WENO7-SR gives the smallest error overall and improves on WENO7-JS and WENO7-Z by 21.3% and 36.3%, respectively. This long-time result is important because errors generated by an inappropriate local weighting rule cannot be removed by shock interactions in linear advection; they accumulate on every traversal of the periodic domain.

4.2. Sod shock tube

The one-dimensional Sod problem is solved on $[0, 1]$ with $N = 100$ finite-volume cells, transmissive boundaries, CFL 0.8, and final time $t = 0.2$. The initial discontinuity is at $x = 0.5$, with left and right states $(\rho, u, p) = (1, 0, 1)$ and $(0.125, 0, 0.1)$. Initial and exact reference cell averages use 15-point Gauss–Legendre quadrature. The symbols therefore represent finite-volume cell averages rather than point samples; the continuous black curve is drawn separately using 10,000 pointwise samples of the exact solution.

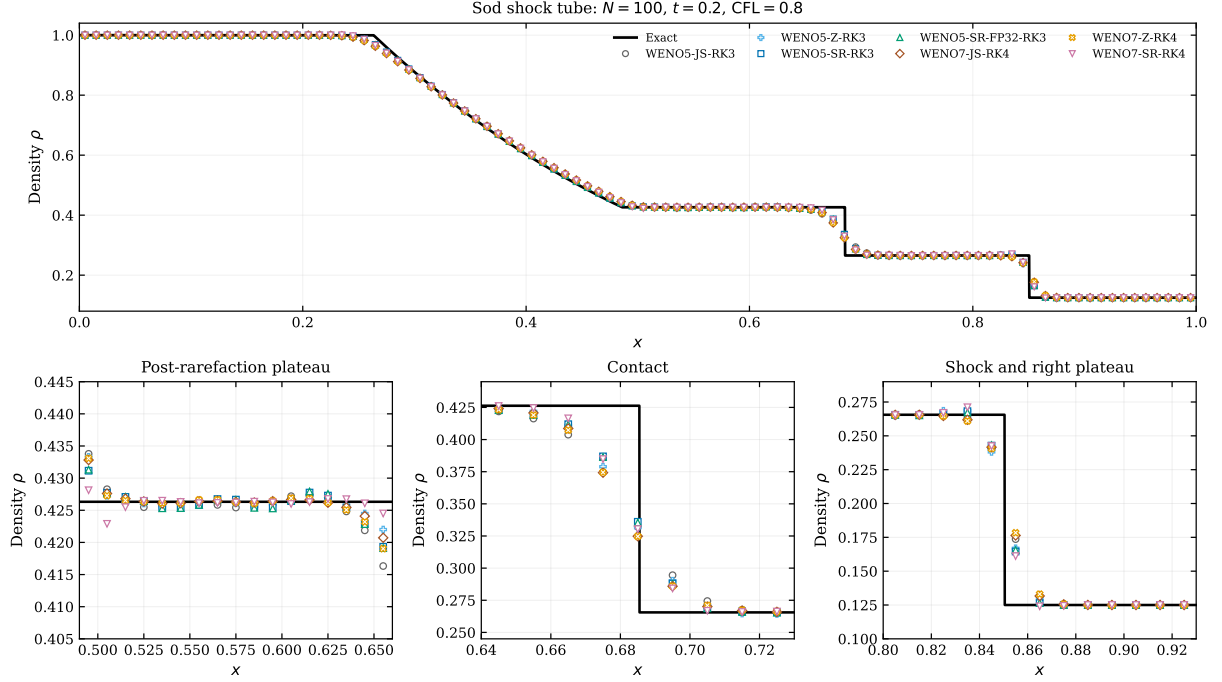


Figure 3: Density for the one-dimensional Sod problem using $N = 100$, CFL 0.8, and $t = 0.2$. Symbols denote finite-volume cell averages, while the black curve is the pointwise exact solution; the lower panels enlarge the post-rarefaction plateau, contact, and shock regions.

Table 3: Density L_1 error for the Sod shock tube using $N = 100$, CFL 0.8, and $t = 0.2$. Numerical and exact states are compared as finite-volume cell averages.

Method	$L_1(\rho)$
WENO5-JS-RK3	4.970×10^{-3}
WENO5-Z-RK3	4.572×10^{-3}
WENO5-SR-RK3	3.806×10^{-3}
WENO5-SR-FP32-RK3	3.887×10^{-3}
WENO7-JS-RK4	4.934×10^{-3}
WENO7-Z-RK4	4.932×10^{-3}
WENO7-SR-RK4	3.997×10^{-3}

The enlarged panels show that the learned WENO5 reconstructions reduce the broadening around the contact and shock relative to WENO5-JS. The FP32 and FP64 WENO5-SR symbols are nearly coincident throughout the domain, indicating that reducing only the MLP precision does not introduce visible oscillatory degradation. WENO7-SR also sharpens the principal waves relative to its classical counterparts. Table 3 confirms the visual comparison: WENO5-SR gives the smallest density error, 23.4% below WENO5-JS and 16.8% below WENO5-Z, while WENO7-SR improves on both WENO7-JS and WENO7-Z by approximately 19%.

4.3. Lax shock tube

The standard Lax shock tube provides a stronger one-dimensional Riemann problem with a pronounced post-shock density plateau. The computational domain is $[-5, 5]$, the initial interface is at $x = 0$, and the primitive states are

$$(\rho, u, p)_L = (0.445, 0.698, 3.528), \quad (\rho, u, p)_R = (0.5, 0, 0.571). \quad (47)$$

All methods use $N = 200$ finite-volume cells, transmissive boundaries, characteristic HLLC fluxes, CFL 0.8, and final time $t = 1.3$. The exact self-similar Riemann solution is integrated separately over each control volume, with the quadrature intervals split at the rarefaction edges, contact, and shock. Consequently, the plotted symbols and all reported errors compare finite-volume averages with exact finite-volume averages rather than with point values.

Table 4: Density L_1 error for the Lax shock tube using $N = 200$, CFL 0.8, and $t = 1.3$. The reference is the exact finite-volume Riemann solution.

Method	$L_1(\rho)$
WENO5-JS-RK3	6.773×10^{-3}
WENO5-Z-RK3	5.813×10^{-3}
WENO5-SR-RK3	4.855×10^{-3}
WENO5-SR-FP32-RK3	4.856×10^{-3}
WENO7-JS-RK4	5.979×10^{-3}
WENO7-Z-RK4	6.091×10^{-3}
WENO7-SR-RK4	4.415×10^{-3}

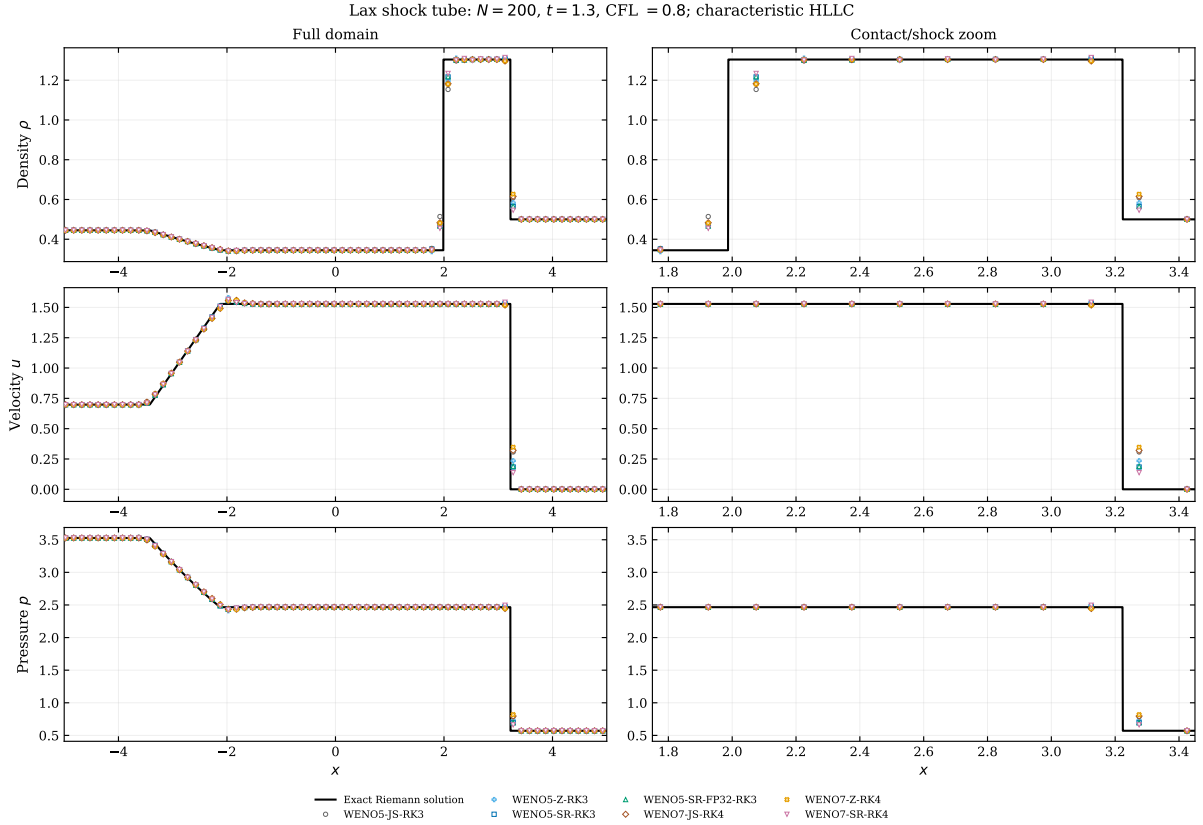


Figure 4: Density, velocity, and pressure for the Lax shock tube at $t = 1.3$ on 200 cells. The left column shows the complete domain and the right column enlarges the contact and shock region. The black curve is the exact self-similar solution; symbols denote numerical finite-volume averages.

All seven calculations reach the prescribed final time without state repair. The learned reconstructions preserve a flatter post-shock plateau and reduce smearing at both the contact and the right-moving shock. Both WENO5-SR variants reduce the density error by more than 28% relative to WENO5-JS and by approximately 16.5% relative to WENO5-Z. WENO7-SR gives the smallest error in Table 4, improving on both WENO7-JS and WENO7-Z by more than 26%. The FP32 and FP64 WENO5-SR profiles remain nearly coincident, including in the enlarged wave region.

4.4. Titarev–Toro shock–entropy-wave interaction

The Titarev–Toro problem [2] provides a substantially longer and more oscillatory one-dimensional Euler test. On $[-5, 5]$, the initial states are

$$(\rho, u, p) = \begin{cases} (1.515695, 0.523346, 1.805), & x < -4.5, \\ (1 + 0.1 \sin(20\pi x), 0, 1), & x \geq -4.5. \end{cases} \quad (48)$$

The calculation uses transmissive boundaries, characteristic HLLC fluxes, CFL 0.8, and final time $t = 5$. Initial cell averages are evaluated by Gaussian quadrature, and errors are measured against a WENO5-JS solution on 10001 cells after conservative restriction.

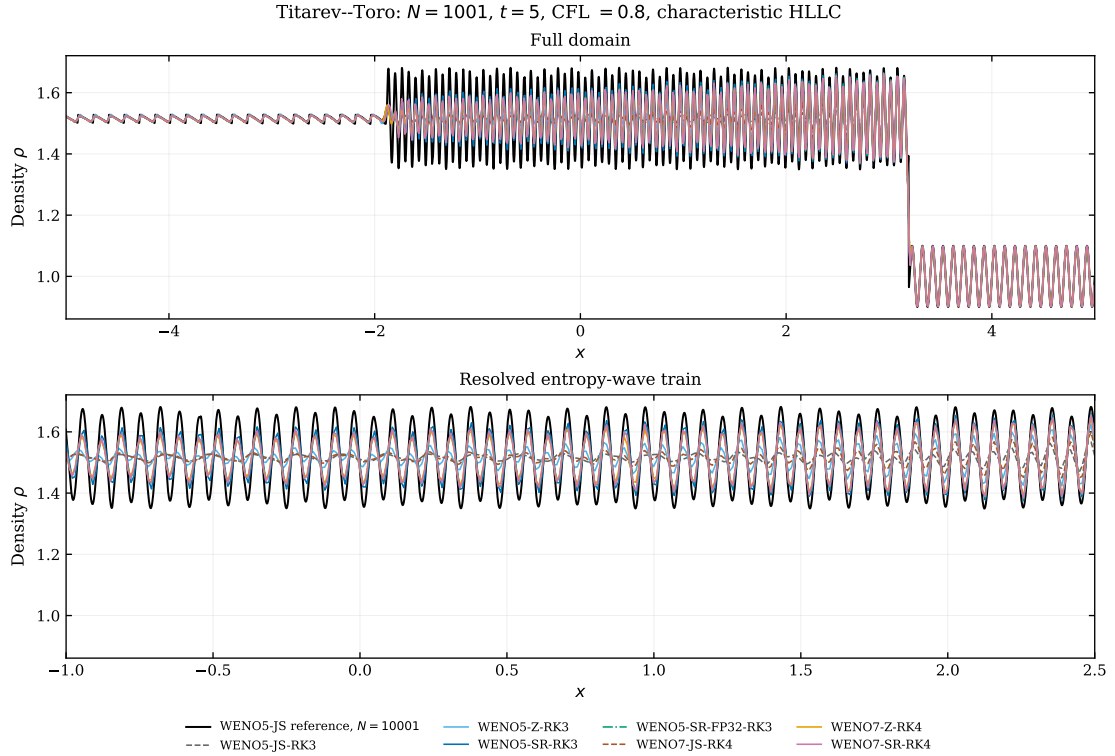


Figure 5: Titarev–Toro density at $t = 5$ using $N = 1001$. The lower panel enlarges the entropy-wave train; all methods use CFL 0.8 and their order-matched SSP Runge–Kutta integrator.

At $N = 1001$, WENO-Z improves on Jiang–Shu by 19.1% for WENO5 and 8.3% for WENO7, confirming that its lower dissipation is useful for transporting the resolved entropy-wave train. The learned reconstructions retain still more of its high-frequency amplitude. WENO5-SR gives the smallest density error at this resolution, reducing the WENO5-JS and WENO5-Z errors by 63.3% and 54.7%, respectively. WENO7-SR gives a comparable error and improves on WENO7-JS and WENO7-Z by 53.1% and 48.9%.

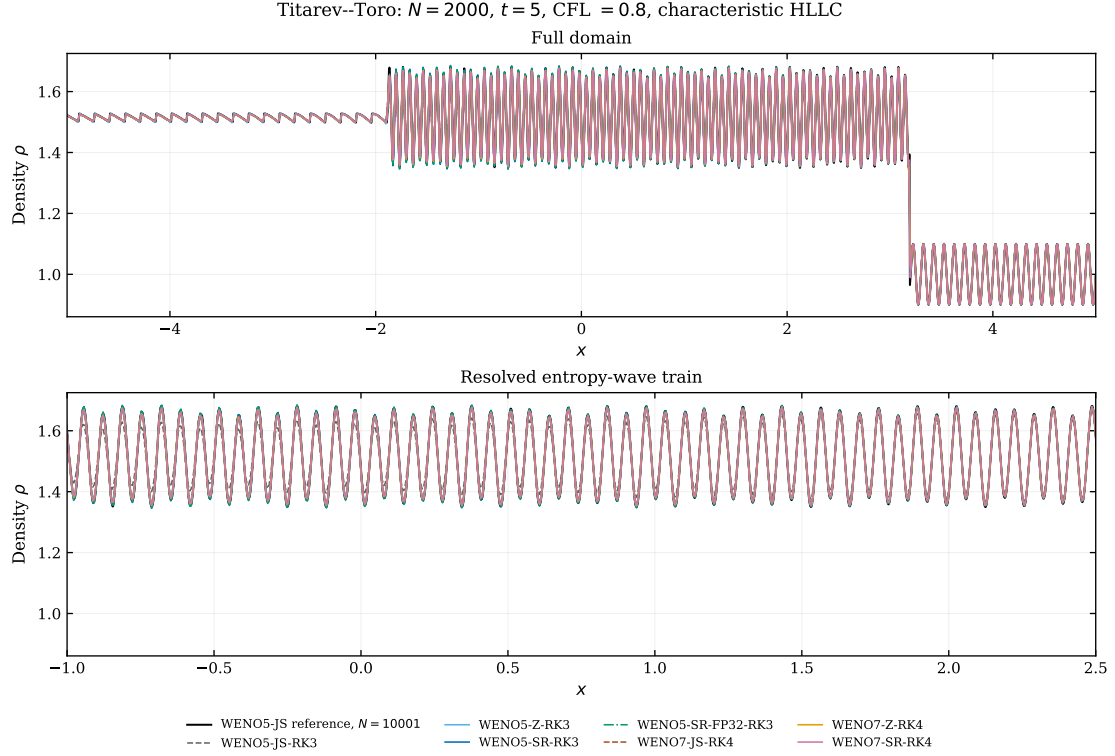


Figure 6: Titarev-Toro density at $t = 5$ using $N = 2000$, with the same numerical configuration and reference as in Fig. 5.

Table 5: Density L_1 error for the Titarev-Toro problem at $t = 5$ and CFL 0.8. Errors use the conservatively restricted WENO5-JS solution on $N = 10001$ as the reference. The lowest value at each resolution is bold.

Method	$N = 1001$	$N = 2000$
WENO5-JS-RK3	4.582×10^{-2}	1.247×10^{-2}
WENO5-Z-RK3	3.709×10^{-2}	4.293×10^{-3}
WENO5-SR-RK3	1.682×10^{-2}	2.121×10^{-3}
WENO5-SR-FP32-RK3	1.961×10^{-2}	1.907×10^{-3}
WENO7-JS-RK4	4.152×10^{-2}	3.199×10^{-3}
WENO7-Z-RK4	3.806×10^{-2}	2.073×10^{-3}
WENO7-SR-RK4	1.945×10^{-2}	1.849×10^{-3}

At $N = 2000$, all seven profiles resolve the entropy-wave train much more sharply, so the visual separation is less pronounced. WENO-Z now reduces the JS error by 65.6% for WENO5 and 35.2% for WENO7, which is a substantial classical improvement. The learned advantage nevertheless remains measurable. WENO7-SR gives the smallest error, followed closely by WENO5-SR-FP32 and WENO7-Z; both WENO5-SR variants remain more accurate than WENO5-Z. The two resolutions therefore show both the coarse-grid benefit of the learned reconstruction and its convergence toward the same resolved wave train under refinement.

4.5. Order verification with an isentropic vortex

We consider the periodic isentropic vortex on $[-10, 10]^2$ with $\gamma = 1.4$, vortex strength $\beta = 5$, and free-stream velocity $(u_\infty, v_\infty) = (1, 1)$. Initial and exact final cell averages are evaluated using tensor-product 15-point Gauss-Legendre quadrature. The solution is advanced to $t = 2$ on 25^2 , 50^2 , 100^2 , and 200^2 meshes. Density errors are measured

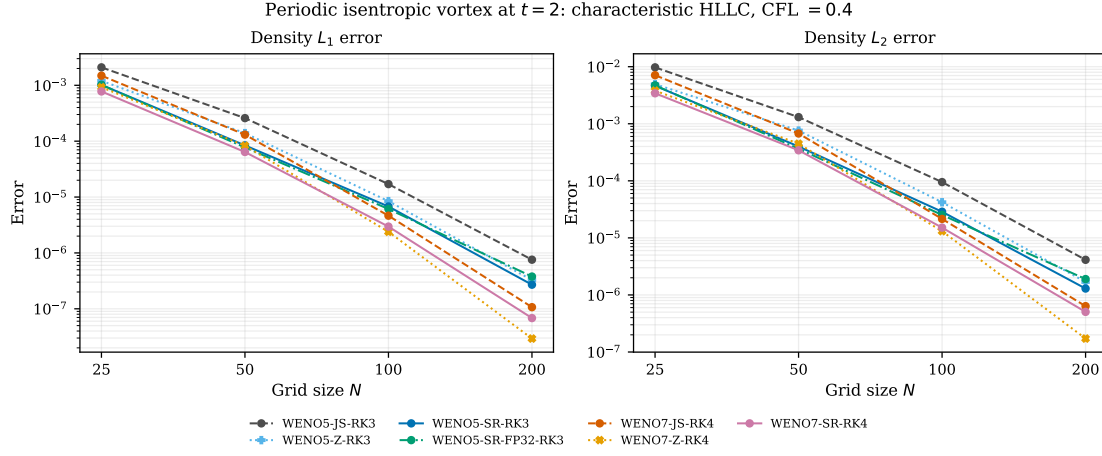


Figure 7: Density-error convergence for the periodic isentropic vortex at $t = 2$ and CFL 0.4. Initial and exact final states are finite-volume cell averages computed with tensor-product 15-point Gauss–Legendre quadrature. The order-matched WENO-Z baselines use $p_Z = r - 1$ and $\varepsilon = h^{r-1}$.

by

$$E_1(\rho) = \frac{1}{N^2} \sum_{i,j} |\rho_{ij} - \rho_{ij}^{\text{ex}}|, \quad E_2(\rho) = \left[\frac{1}{N^2} \sum_{i,j} (\rho_{ij} - \rho_{ij}^{\text{ex}})^2 \right]^{1/2}. \quad (49)$$

Table 6: Density errors and observed orders for the periodic isentropic vortex at $t = 2$ and CFL 0.4.

Method	N	$L_1(\rho)$	order	$L_2(\rho)$	order
WENO5-JS-RK3	25	2.095e-03	—	9.741e-03	—
WENO5-JS-RK3	50	2.585e-04	3.019	1.307e-03	2.898
WENO5-JS-RK3	100	1.705e-05	3.922	9.508e-05	3.781
WENO5-JS-RK3	200	7.565e-07	4.494	4.128e-06	4.526
WENO5-Z-RK3	25	1.204e-03	—	4.939e-03	—
WENO5-Z-RK3	50	1.394e-04	3.111	7.597e-04	2.701
WENO5-Z-RK3	100	8.441e-06	4.045	4.213e-05	4.172
WENO5-Z-RK3	200	3.271e-07	4.690	1.688e-06	4.641
WENO5-SR-RK3	25	1.017e-03	—	4.641e-03	—
WENO5-SR-RK3	50	8.401e-05	3.597	3.996e-04	3.538
WENO5-SR-RK3	100	6.778e-06	3.632	2.875e-05	3.797
WENO5-SR-RK3	200	2.701e-07	4.649	1.295e-06	4.472
WENO5-SR-FP32-RK3	25	1.017e-03	—	4.715e-03	—
WENO5-SR-FP32-RK3	50	7.741e-05	3.715	3.647e-04	3.692
WENO5-SR-FP32-RK3	100	6.118e-06	3.661	2.522e-05	3.854
WENO5-SR-FP32-RK3	200	3.777e-07	4.018	1.899e-06	3.731
WENO7-JS-RK4	25	1.487e-03	—	7.105e-03	—
WENO7-JS-RK4	50	1.308e-04	3.507	6.750e-04	3.396
WENO7-JS-RK4	100	4.642e-06	4.817	2.149e-05	4.973
WENO7-JS-RK4	200	1.070e-07	5.439	6.362e-07	5.078
WENO7-Z-RK4	25	9.247e-04	—	3.808e-03	—
WENO7-Z-RK4	50	8.301e-05	3.478	4.474e-04	3.089
WENO7-Z-RK4	100	2.390e-06	5.118	1.311e-05	5.093
WENO7-Z-RK4	200	2.933e-08	6.348	1.720e-07	6.252
WENO7-SR-RK4	25	7.766e-04	—	3.424e-03	—
WENO7-SR-RK4	50	6.400e-05	3.601	3.430e-04	3.320
WENO7-SR-RK4	100	2.962e-06	4.433	1.514e-05	4.502
WENO7-SR-RK4	200	6.826e-08	5.440	5.048e-07	4.907

All seven methods converge regularly over the four meshes. The distance-balanced WENO5-SR model reaches observed orders 4.65 and 4.47 in E_1 and E_2 , respectively, between the two finest meshes. At $N = 200$ its errors, 2.701×10^{-7} and 1.295×10^{-6} , are 64.3% and 68.6% smaller than those of WENO5-JS, and remain 17.4% and 23.3% below WENO5-Z. WENO7-Z gives the smallest fine-grid smooth-solution errors, $E_1 = 2.933 \times 10^{-8}$ and

$E_2 = 1.720 \times 10^{-7}$, while WENO7-SR retains regular high-order convergence and remains more accurate than WENO7-JS. The FP32 WENO5-SR network is likewise stable and convergent, reaching $E_1 = 3.777 \times 10^{-7}$ and $E_2 = 1.899 \times 10^{-6}$ on the finest mesh. Thus, the learned shock-capturing closure does not sacrifice the high-order finite-volume limit, while WENO-Z supplies a particularly strong classical baseline for this globally smooth test.

This smooth benchmark also clarifies the role of the WENO-Z comparison. Its global smoothness correction accelerates recovery of the optimal linear weights near smooth critical points, and the original analyses demonstrate the resulting accuracy improvement [6, 7]. Our GSTE, Titarev–Toro, and isentropic-vortex results confirm that this mechanism can produce substantial gains in smooth transport and in flows dominated by a resolved wave train. Castro et al. [7] also note that, once shocks dominate the global error, critical-point order alone does not explain improved shock resolution; the nonlinear dissipation selected near discontinuities becomes decisive. Consistently, the shocked-flow tests below show that the critical-point advantage does not always translate into superior resolution of under-resolved wave interactions. A global smoothness correction does not provide the same state-conditioned discrimination among competing substencils as the learned map. WENO-SR preserves the designed convergence order while learning a more state-adaptive balance among dissipation, sharpness, and nonlinear stability over the evolved solution distribution.

4.6. Two-dimensional Riemann problems

We next consider configurations C.3–C.6 from the two-dimensional Riemann test family used in earlier neural-WENO studies [4]. The initial discontinuities are aligned with $x = 0.5$ and $y = 0.5$ on $[0, 1]^2$. States 1–4 in Table 7 occupy the upper-right, upper-left, lower-left, and lower-right quadrants, respectively. All initial finite-volume averages are evaluated with tensor-product 15-point Gauss–Legendre quadrature. Each calculation uses a 400^2 mesh, outflow boundaries, CFL 0.4, characteristic HLLC fluxes, and no ENO cutoff.

Table 7: Primitive states (ρ, u, v, p) and output times for the two-dimensional Riemann configurations.

Configuration	State 1	State 2	State 3	State 4	t_{out}
C.3	(1.5, 0, 0, 1.5)	(0.5323, 1.206, 0, 0.3)	(0.138, 1.206, 1.206, 0.029)	(0.5323, 0, 1.206, 0.3)	0.50
C.4	(1.1, 0, 0, 1.1)	(0.5065, 0.8939, 0, 0.35)	(1.1, 0.8939, 0.8939, 1.1)	(0.5065, 0, 0.8939, 0.35)	0.25
C.5	(1, −0.75, −0.5, 1)	(2, −0.75, 0.5, 1)	(1, 0.75, 0.5, 1)	(3, 0.75, −0.5, 1)	0.23
C.6	(1, 0.75, −0.5, 1)	(2, 0.75, 0.5, 1)	(1, −0.75, 0.5, 1)	(3, −0.75, −0.5, 1)	0.30

All six methods complete all four calculations without non-finite states or loss of density or pressure positivity. Presenting every method at the same mesh resolution and with common plotting limits permits a direct qualitative comparison of contact resolution, shear-layer development, and preservation of the large-scale wave geometry without method-dependent visual rescaling. Within each configuration, the density contours are uniformly spaced over a fixed range and exactly the same levels are used for all six methods. The range and number of contours may differ between configurations to accommodate their distinct density ranges. Pressure is rendered with the same colour map throughout, while every panel belonging to a given configuration shares one pressure normalisation. The respective $(\rho_{\min}, \rho_{\max}, N_\rho)$ contour specifications are (0.16, 1.71, 32), (0.64, 1.92, 19), (1.10, 3.90, 21), and (0.54, 2.94, 25) for C.3–C.6. The corresponding increments are 0.05, 0.0711, 0.14, and 0.10, respectively; both endpoints are included. For C.4 only, the plotting coordinates are translated by (0.5, 0.5) so that the initial quadrant intersection is at the origin and the displayed domain is $[-0.5, 0.5]^2$. This translation does not alter the numerical solution.

Two-dimensional Riemann configuration C.3 at $t = 0.50$

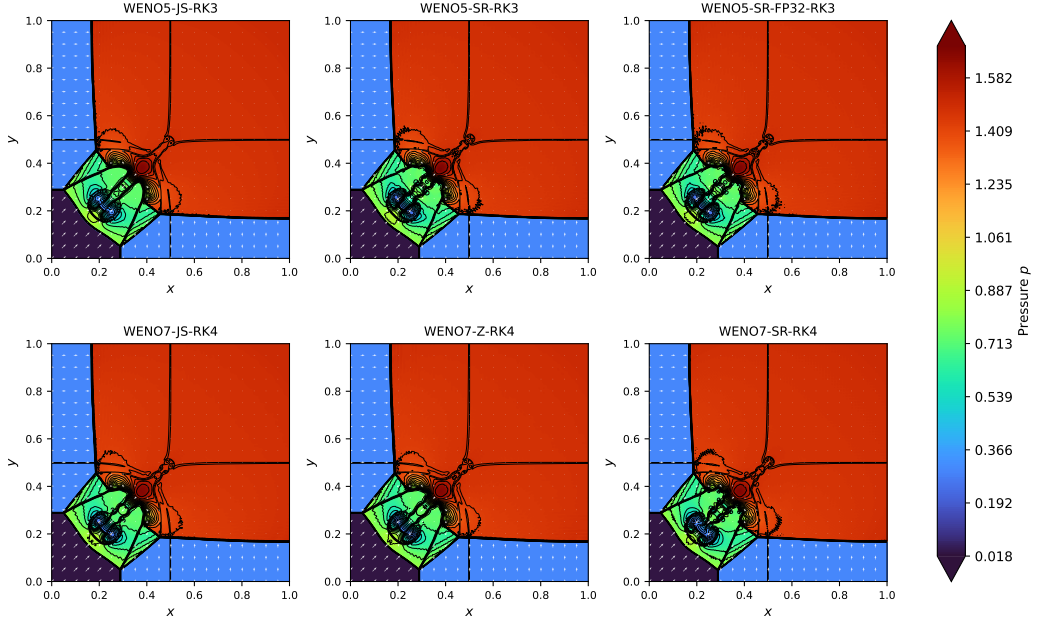


Figure 8: Configuration C.3 (q400) at $t = 0.50$ on a 400^2 mesh. Pressure is shown in colour, black curves are density contours, and white arrows indicate velocity; all six panels share the same colour scale and contour levels. WENO7-Z uses $p_Z = 3$ and $\varepsilon = h^3$.

Two-dimensional Riemann configuration C.4 at $t = 0.25$

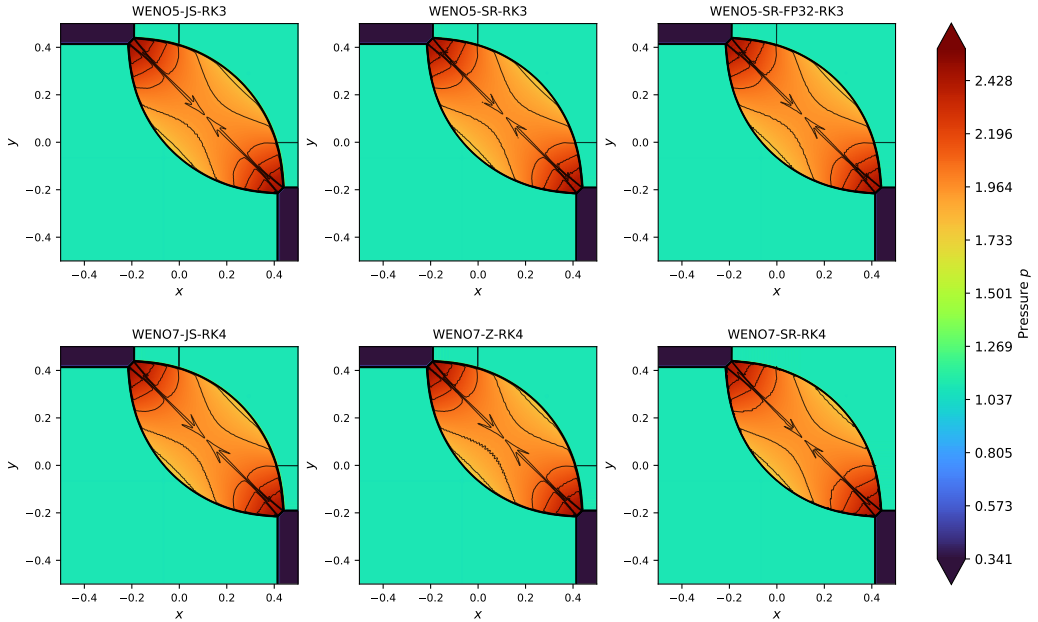


Figure 9: Configuration C.4 at $t = 0.25$ on a 400^2 mesh. Pressure is shown in colour and black curves are 19 equally spaced density contours over $0.64 \leq \rho \leq 1.92$. Coordinates are centred at the initial discontinuity, and all method panels share the same pressure colour scale and density-contour levels.

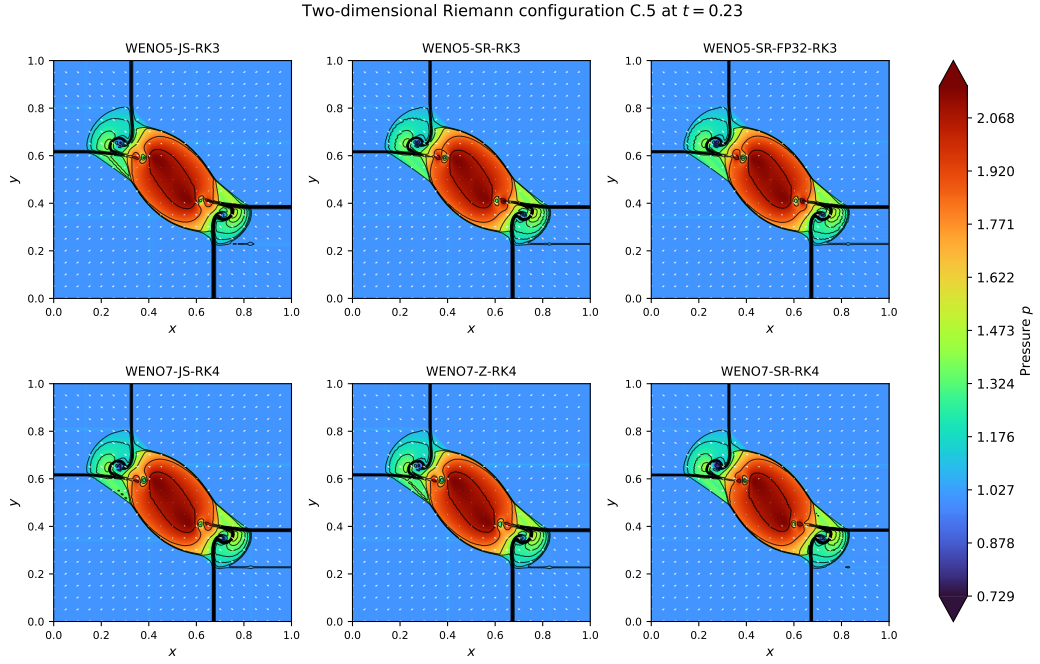


Figure 10: Configuration C.5 at $t = 0.23$ on a 400^2 mesh. All method panels share one pressure colour scale and one set of density-contour levels.

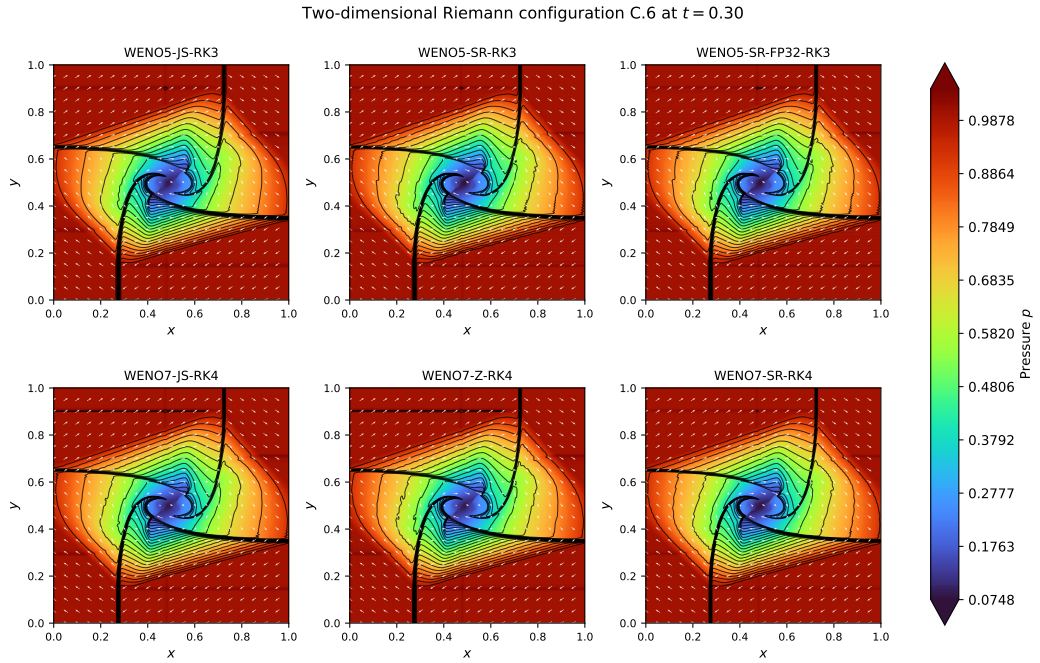


Figure 11: Configuration C.6 at $t = 0.30$ on a 400^2 mesh. All method panels share one pressure colour scale and one set of density-contour levels.

4.7. Double Mach reflection

A 1200×300 double Mach reflection calculation provides an additional out-of-distribution robustness test for all three selected WENO-SR models and their order-matched JS and Z baselines. Every calculation uses characteristic reconstruction, HLLC, CFL 0.4, 15-point finite-volume initialisation, and no ENO cutoff or state repair. WENO5-JS, WENO5-Z, WENO5-SR-FP64, and WENO5-SR-FP32 reach $t = 0.2$ in 3203, 3215, 3234, and 3251 SSPRK3 steps, respectively. WENO7-JS, WENO7-Z, and WENO7-SR reach the same time in 3178, 3177, and 3256 fourth-order SSP Runge–Kutta steps. All seven calculations remain free of non-finite values and negative density or pressure.

The learned solutions preserve the baseline large-scale shock geometry while resolving additional structure near the jet and reflected-shock interaction. The FP32 WENO5 network reproduces the same principal wave pattern as its FP64 counterpart, with no precision-specific failure. Figures 12 and 13 pair each complete field with the same enlargement of the slip-line and wall-jet interaction. All seven solutions use one density colour scale and the same 29 black density contours, uniformly spaced over the common plotted density range. Consequently, the additional closed roll-ups in the WENO-SR panels reflect differences in the computed fields rather than method-dependent plotting choices. In particular, WENO7-SR resolves a longer hierarchy of coherent secondary vortices along the slip line and near the wall than either order-matched classical weighting.

For qualitative context, Fig. 14 reproduces the 1600×400 row of the published high-order WENO-Z comparison of Castro et al. [7]. That row contains fifth-, ninth-, and eleventh-order WENO-Z results with $p_Z = r - 1$. Although differences in spatial order, mesh, flux formulation, and plotting convention preclude a quantitative cross-study error comparison, the roll-up complexity resolved by WENO7-SR on the coarser 1200×300 mesh is qualitatively comparable to that visible in the published eleventh-order WENO-Z result at 1600×400 .

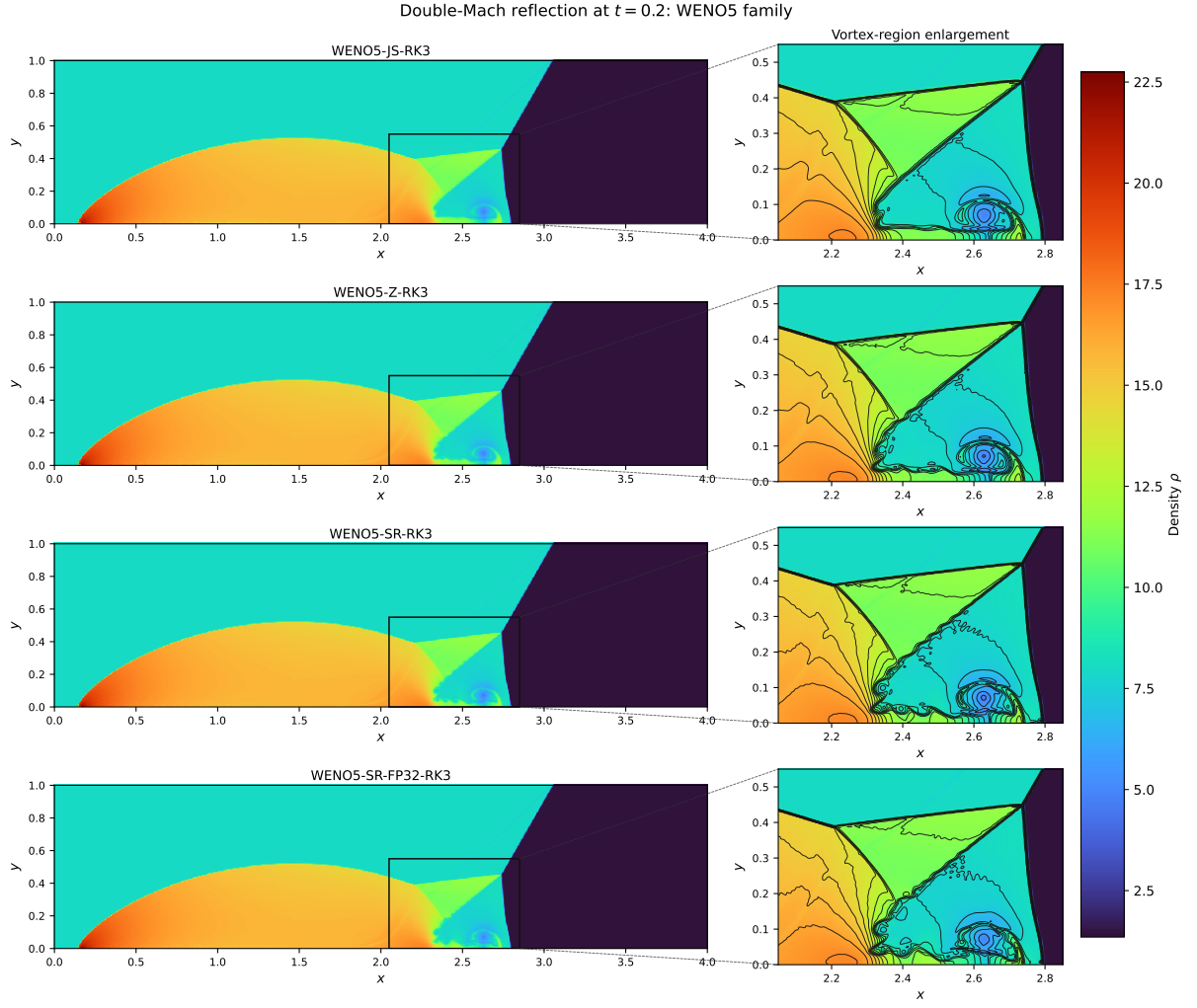


Figure 12: Double Mach reflection at $t = 0.2$ for the WENO5 family on a 1200×300 mesh. Each row shows the complete density field (left) and the enlargement $2.05 \leq x \leq 2.85$, $0 \leq y \leq 0.55$ (right). Colour limits and 29 equally spaced density-contour levels are common to all methods and are identical to those in Fig. 13.

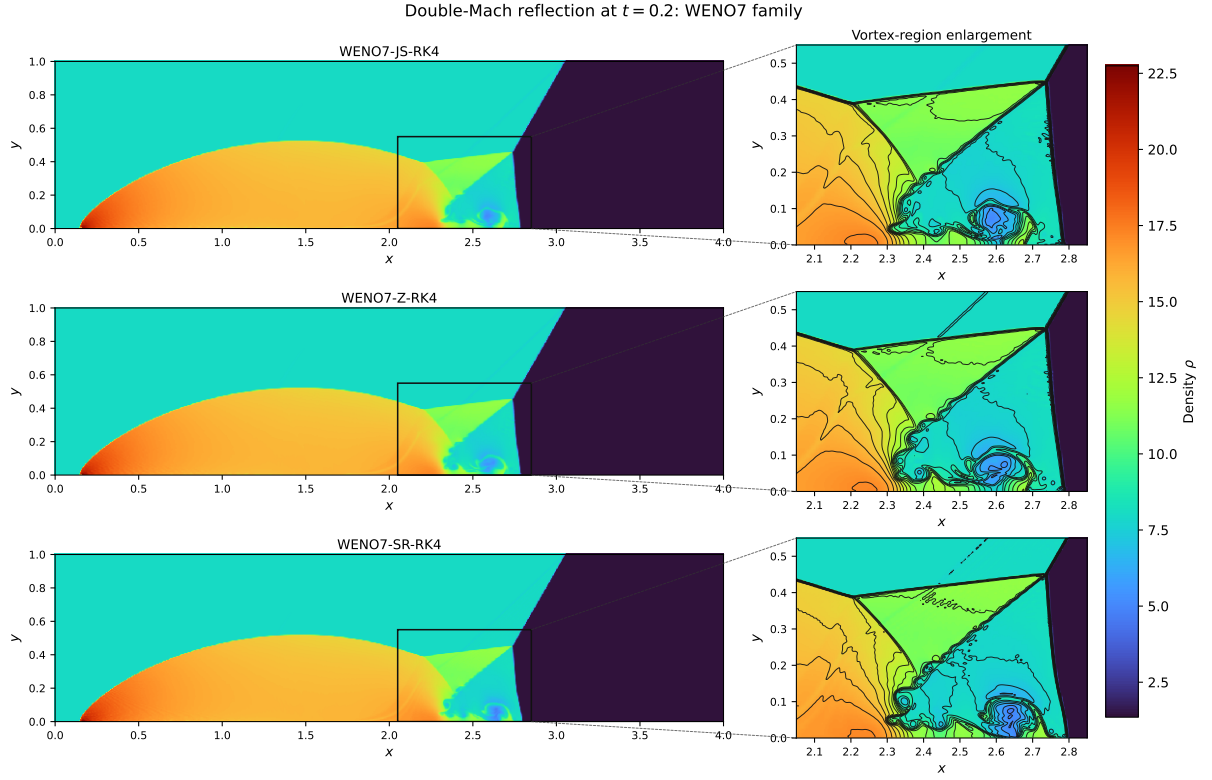


Figure 13: Double Mach reflection at $t = 0.2$ for the WENO7 family on a 1200×300 mesh. The plotting window, density colour scale, and equally spaced contour levels are identical to those in Fig. 12.

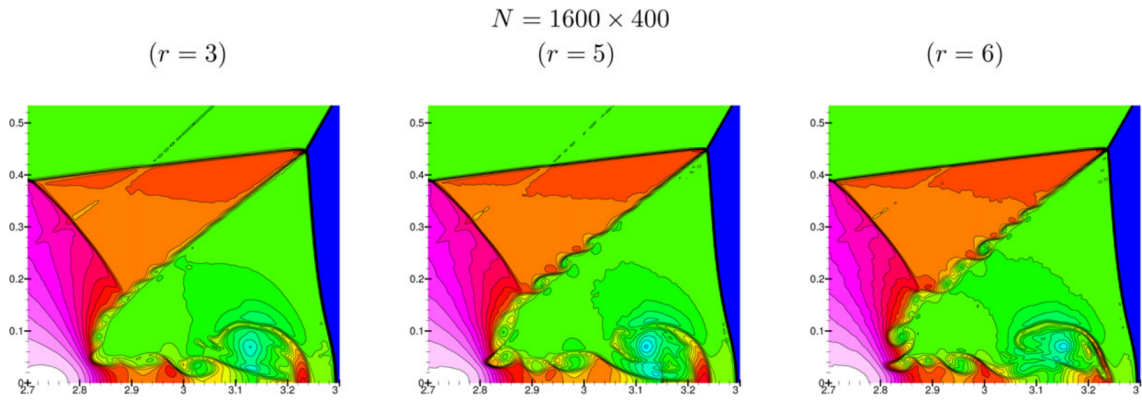


Figure 14: Published 1600×400 density-contour comparison for fifth- ($r = 3$), ninth- ($r = 5$), and eleventh-order ($r = 6$) WENO-Z at $t = 0.2$, reproduced from Castro et al. [7] for qualitative context.

4.8. Planar shock–helium-bubble interaction

We finally test the selected reconstructions on planar shock–helium-bubble interactions at incident shock Mach numbers $M_s = 1.22$ and $M_s = 3.0$. The domain is $[0, 0.225] \times [0, 0.089]$, and the helium bubble has radius 0.025 and centre (0.035, 0.0445). Every calculation uses 15-point finite-volume initialisation, transmissive boundaries, characteristic HLLC fluxes, CFL 0.4, and no ENO cutoff. The $M_s = 1.22$ solution is advanced to $t = 6.0 \times 10^{-4}$, whereas the $M_s = 3.0$ solution is shown at $t = 1.0 \times 10^{-4}$. All seven methods are computed on a 1000×396 mesh. An independently computed 2000×791 WENO7-JS-RK4 solution is included as the reference in each comparison.

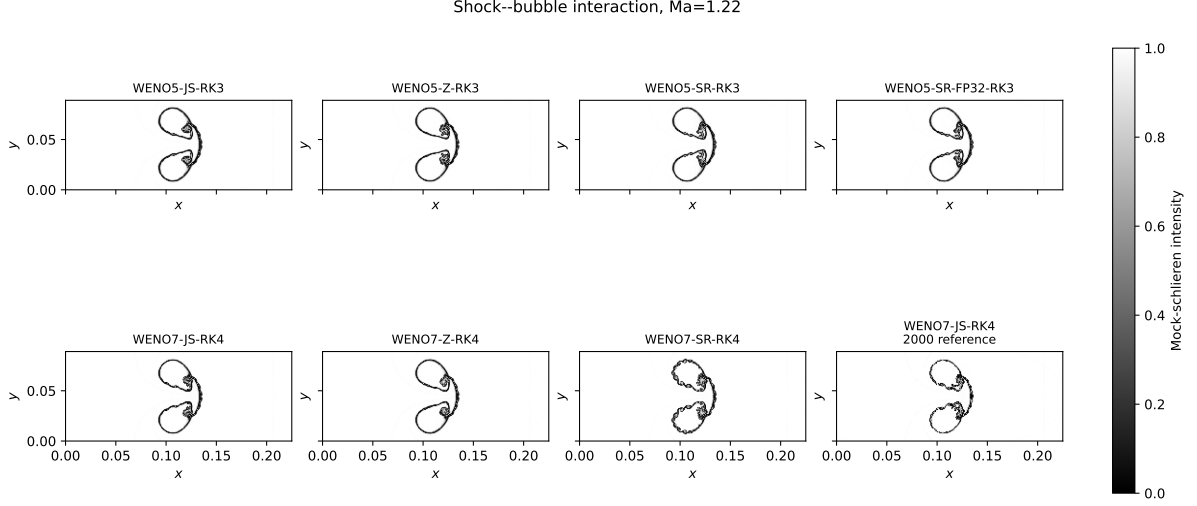


Figure 15: Full-domain mock-schlieren comparison for the $M_s = 1.22$ shock–helium-bubble interaction at $t = 6.0 \times 10^{-4}$. The seven learned or classical solutions use 1000×396 cells; the lower-right panel is the 2000×791 WENO7-JS-RK4 reference. All panels share one normalisation.

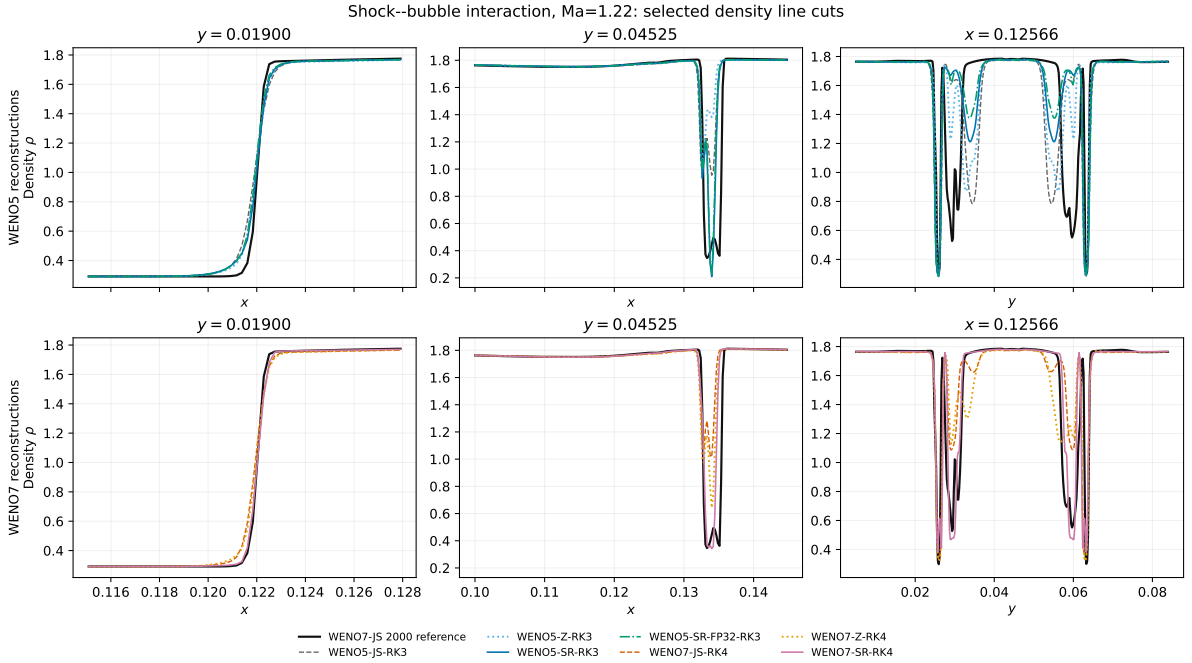


Figure 16: Selected density profiles for the $M_s = 1.22$ interaction. The cuts are (a) $y = 0.01900$ over $0.115 \leq x \leq 0.128$, (b) $y = 0.04525$ over $0.100 \leq x \leq 0.145$, and (c) $x = 0.12566$ over $0.005 \leq y \leq 0.084$. The fine-grid reference is conservatively restricted before comparison.

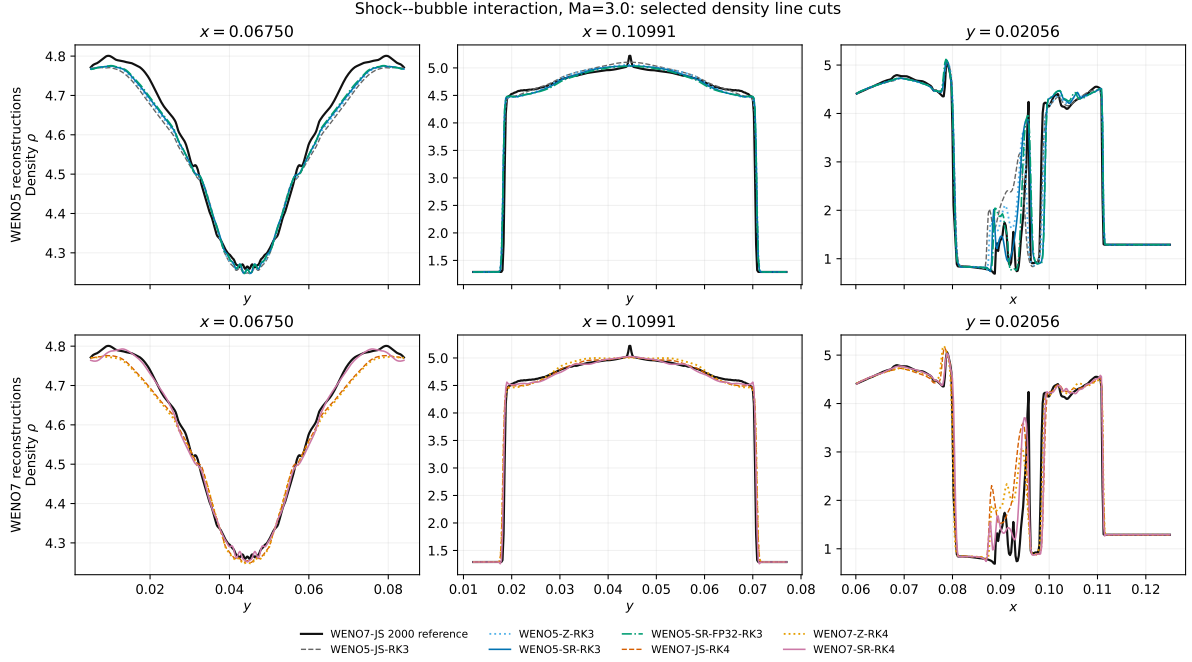


Figure 18: Selected density profiles for the $M_s = 3.0$ interaction. The cuts are (a) $x = 0.06750$ over $0.005 \leq y \leq 0.084$, (b) $x = 0.10991$ over $0.012 \leq y \leq 0.077$, and (c) $y = 0.02056$ over $0.060 \leq x \leq 0.125$.

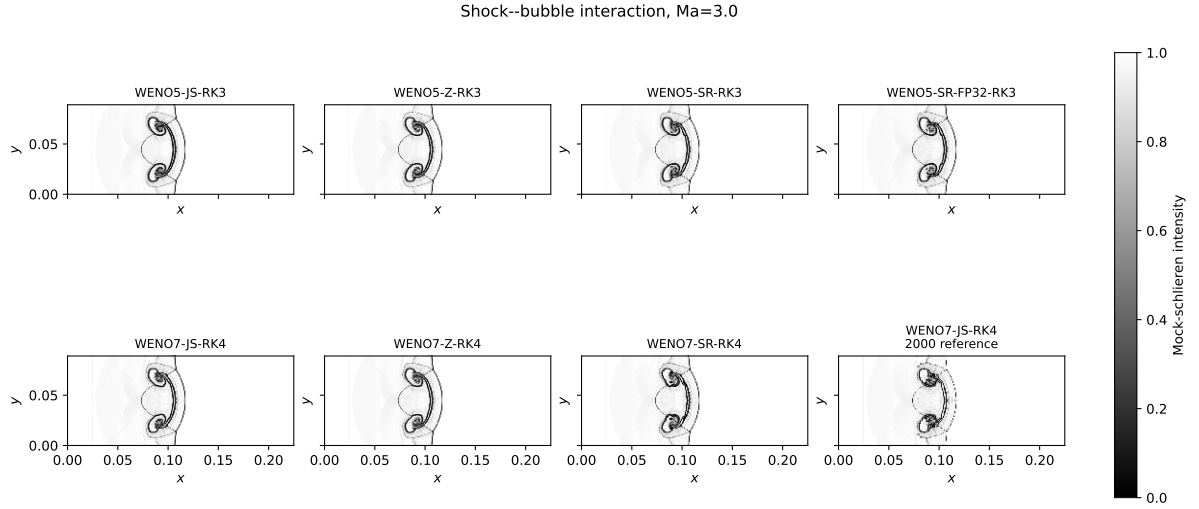


Figure 17: Full-domain mock-schlieren comparison for the $M_s = 3.0$ shock-helium-bubble interaction at $t = 1.0 \times 10^{-4}$. The seven 1000×396 solutions and the 2000×791 WENO7-JS-RK4 reference share one normalisation.

For both Mach numbers, the principal upper and lower vortex cores reconstructed by WENO7-SR occur at nearly the same streamwise and transverse locations as in the 2000×791 reference. The rolled-up interface and the connecting compressed layer are also more sharply resolved than in the order-matched 1000×396 WENO7-JS solution. This agreement is obtained without using the shock-bubble configuration in training. The WENO5-SR solutions likewise reduce diffusion relative to WENO5-JS, while the FP32 network preserves the same large-scale interaction geometry as the FP64 model.

Table 8: Local density L_1 errors for the selected shock–bubble profiles, measured against the conservatively restricted 2000×791 WENO7-JS–RK4 reference. The integration intervals are those stated in Figs. 16 and 18. The smallest error in each row is shown in bold.

M_s	Cut	Interval	W5-JS	W5-Z	W5-SR64	W5-SR32	W7-JS	W7-Z	W7-SR
1.22	$y = 0.01900$	$x \in [0.115, 0.128]$	3.0584×10^{-2}	2.4323×10^{-2}	2.1775×10^{-2}	2.2924×10^{-2}	2.1561×10^{-2}	2.0974×10^{-2}	7.5029×10^{-3}
1.22	$y = 0.04525$	$x \in [0.100, 0.145]$	6.2331×10^{-2}	7.1680×10^{-2}	4.7419×10^{-2}	4.8432×10^{-2}	6.1765×10^{-2}	5.2943×10^{-2}	3.0217×10^{-2}
1.22	$x = 0.12566$	$y \in [0.005, 0.084]$	2.1462×10^{-1}	1.7144×10^{-1}	1.7108×10^{-1}	1.5679×10^{-1}	1.0450×10^{-1}	1.1924×10^{-1}	6.7429×10^{-2}
3.0	$x = 0.06750$	$y \in [0.005, 0.084]$	2.9790×10^{-2}	2.4894×10^{-2}	2.2424×10^{-2}	2.1160×10^{-2}	2.0824×10^{-2}	2.3610×10^{-2}	8.9685×10^{-3}
3.0	$x = 0.10991$	$y \in [0.012, 0.077]$	7.0941×10^{-2}	7.1270×10^{-2}	6.5105×10^{-2}	6.4304×10^{-2}	6.0780×10^{-2}	7.5344×10^{-2}	4.5593×10^{-2}
3.0	$y = 0.02056$	$x \in [0.060, 0.125]$	2.2472×10^{-1}	1.9020×10^{-1}	1.7474×10^{-1}	1.6221×10^{-1}	2.1787×10^{-1}	1.9172×10^{-1}	1.4844×10^{-1}

Table 8 quantifies the selected profiles. For each cut, the 2000×791 reference is conservatively restricted to the overlapping coarse cells before the local density L_1 norm is formed. WENO7-SR has the smallest error among all seven 1000×396 methods on all six cuts. Relative to WENO7-JS, its error reductions are 65.2%, 51.1%, and 35.5% for $M_s = 1.22$, and 56.9%, 25.0%, and 31.9% for $M_s = 3.0$. On all three revised cuts, both WENO5-SR implementations also have smaller density L_1 errors than WENO5-Z, while WENO7-SR remains more accurate than every WENO5 result.

4.9. Three-dimensional single-material shock–bubble interaction

We next examine whether the learned WENO5 closure transfers to a genuinely three-dimensional compressible-flow calculation. The benchmark consists of a planar Mach-3 shock moving in the positive x -direction and interacting with a spherical low-density bubble. As in the corresponding high-resolution study, the bubble is treated as a single-material density interface: the same ideal-gas equation of state with $\gamma = 1.4$ is used throughout the domain. The computational domain and geometric parameters are

$$\Omega = [0, 0.225] \times [0, 0.089] \times [0, 0.089], \quad x_s = 0.005, \quad (x_c, y_c, z_c) = (0.035, 0.0445, 0.0445), \quad R = 0.025. \quad (50)$$

Writing the primitive state as $\mathbf{V} = (\rho, u, v, w, p)$, the states to the left and right of the incident shock are

$$\mathbf{V}(x, 0) = \begin{cases} (4.975714, 736.911, 0, 0, 1.047025 \times 10^6), & x < x_s, \\ (1.29, 0, 0, 0, 101325), & x \geq x_s, \end{cases} \quad (51)$$

and the state inside the sphere $(x - x_c)^2 + (y - y_c)^2 + (z - z_c)^2 \leq R^2$ is replaced by $\mathbf{V}_b = (0.214, 0, 0, 0, 101325)$.

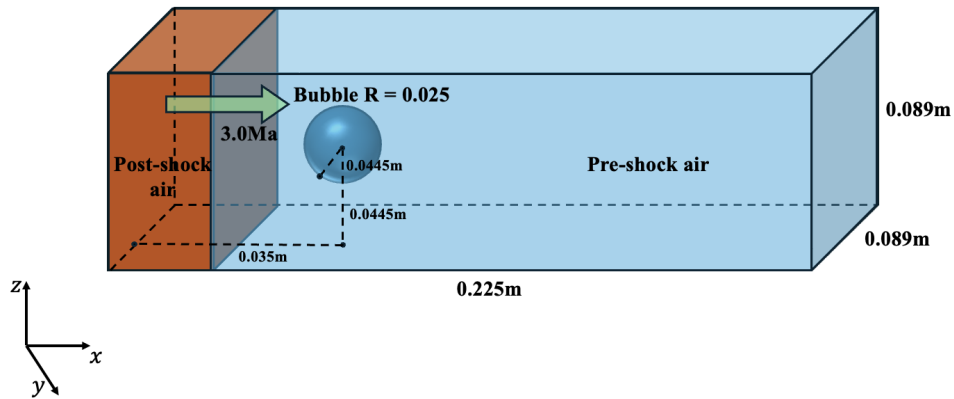


Figure 19: Initial configuration of the three-dimensional single-material shock–bubble benchmark. A planar Mach-3 shock initially located at $x = x_s$ propagates through air and impacts a spherical light-density region of radius R .

All cell averages are initialised with a tensor-product $3 \times 3 \times 3$ Gauss rule. The principal comparison uses a $224 \times 88 \times 88$ Cartesian mesh, CFL 0.25, transmissive boundaries, characteristic reconstruction, the EVILIN Riemann solver, and no ENO cutoff. EVILIN is retained here because the high-resolution WENO7–ADER4 reference uses the same flux, so reconstruction and resolution remain the only intended numerical differences. All solutions are advanced to $t = 10^{-4}$. Seven methods are considered at this resolution: the JS, Z, and SR variants of WENO5 coupled to SSPRK3; the mixed-precision WENO5-SR model; and the JS, Z, and SR variants of WENO7 coupled to the four-stage, fourth-order downwind TVD Runge–Kutta method. The selected FP64 WENO5-SR and WENO7-SR networks are taken at training steps 12 250 and 16 750, respectively; the mixed-precision WENO5-SR network is taken at step 16 500. Every learned reconstruction uses reflection-symmetrised inference. In the mixed-precision variant, only the MLP parameters and forward pass use FP32; the finite-volume state, characteristic transformations, Riemann solver, fluxes, and Runge–Kutta update remain in FP64. Classical WENO5-JS–SSPRK3 and WENO7-JS–RK4 calculations on a $448 \times 176 \times 176$ mesh are also included to show the effect of uniform mesh refinement.

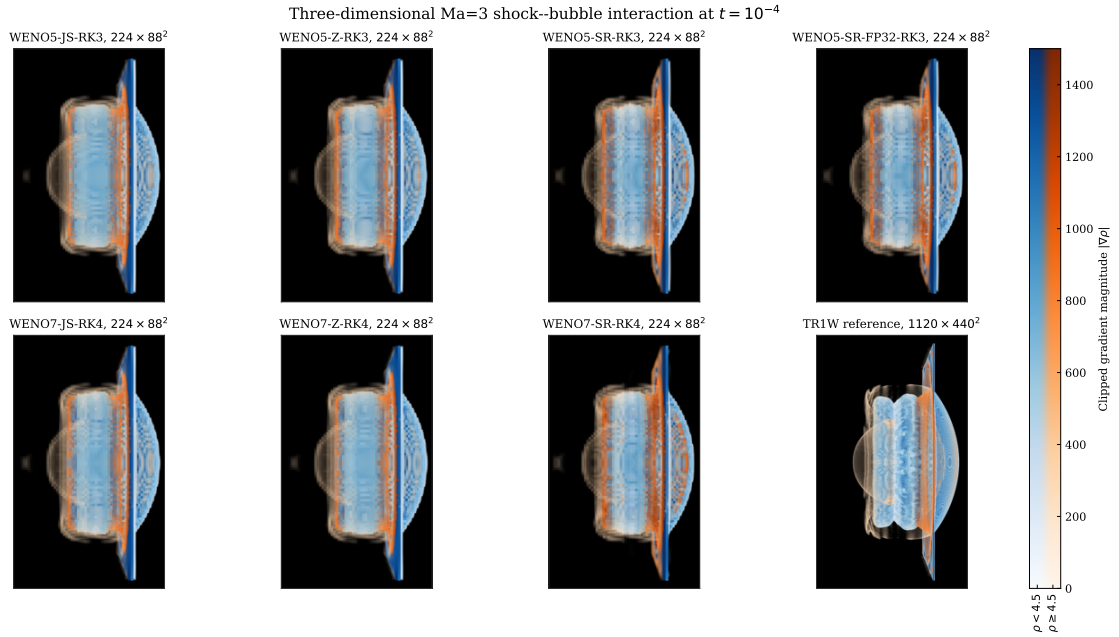


Figure 20: Volume rendering of the three-dimensional Mach-3 shock–bubble interaction at $t = 10^{-4}$. The seven methods use the $224 \times 88 \times 88$ mesh, whereas the last panel is the $1120 \times 440 \times 440$ WENO7–ADER4 reference. A common transfer function and clipped density–gradient range are used in all panels.

The FP64 and MLP-FP32 WENO5-SR solutions in Figs. 20 and 21 are visually almost indistinguishable. Their differences from WENO5-JS are concentrated around the rolled-up bubble interface and the adjacent multiscale interaction region. Over the resolved smooth portions of the flow, all three reconstructions approach the same fifth-order smooth limit and consequently produce nearly identical fields. This localisation is the intended behaviour of WENO-SR: the learned closure modifies nonlinear stencil selection where under-resolution is important, while retaining the classical high-order reconstruction elsewhere.

Three-dimensional Ma=3 shock-bubble, central z plane, $t = 10^{-4}$

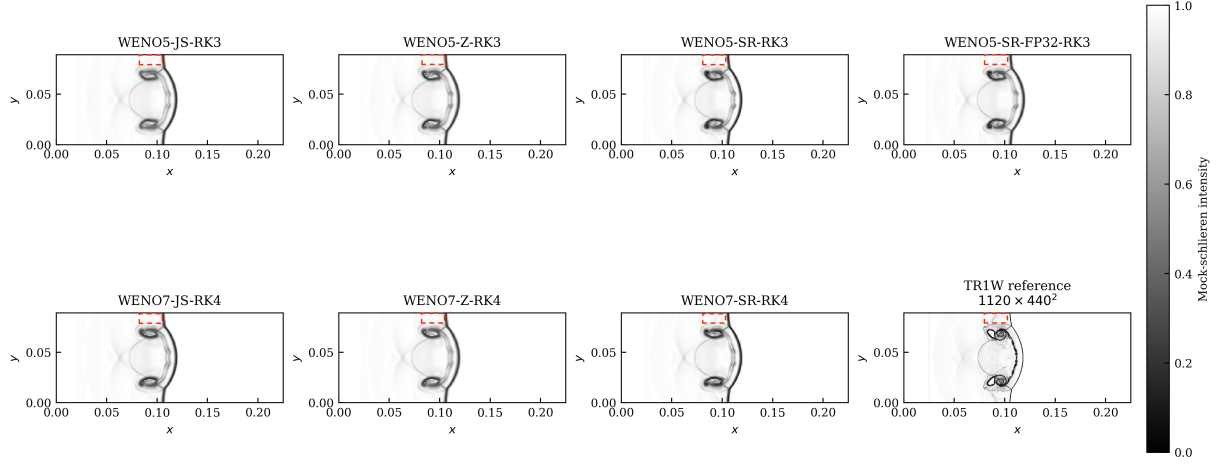


Figure 21: Central-z-plane mock-schlieren fields for the same three-dimensional interaction, with a common intensity scale. The red boxes identify the same upper shock–interface interaction region in every panel. Relative to the order-matched JS and Z baselines, the WENO-SR solutions retain a sharper transmitted-shock trace and reproduce the WENO7–ADER4 reference roll-up topology more faithfully in this region.

For a quantitative local comparison, density profiles are extracted along the y -direction near the central plane $z = 0.0445$ at the nominal locations $x = 0.085, 0.097$, and 0.114 . These cuts intersect, respectively, the rolled-up vortex system, the strongly deformed interaction core, and the leading bubble front. A WENO7–ADER4 computation on a $1120 \times 440 \times 440$ mesh at $t = 10^{-4}$ provides the high-resolution reference. The profiles include all seven methods on the $224 \times 88 \times 88$ mesh and both classical methods on the $448 \times 176 \times 176$ mesh. For the even 2:1 refinement, the two fine-grid samples symmetric about the selected coarse-grid cell centre are averaged; for the aligned 5:1 reference, the coincident fine-grid cell centre is used. This removes an otherwise arbitrary half-cell offset from the cross-resolution comparison.

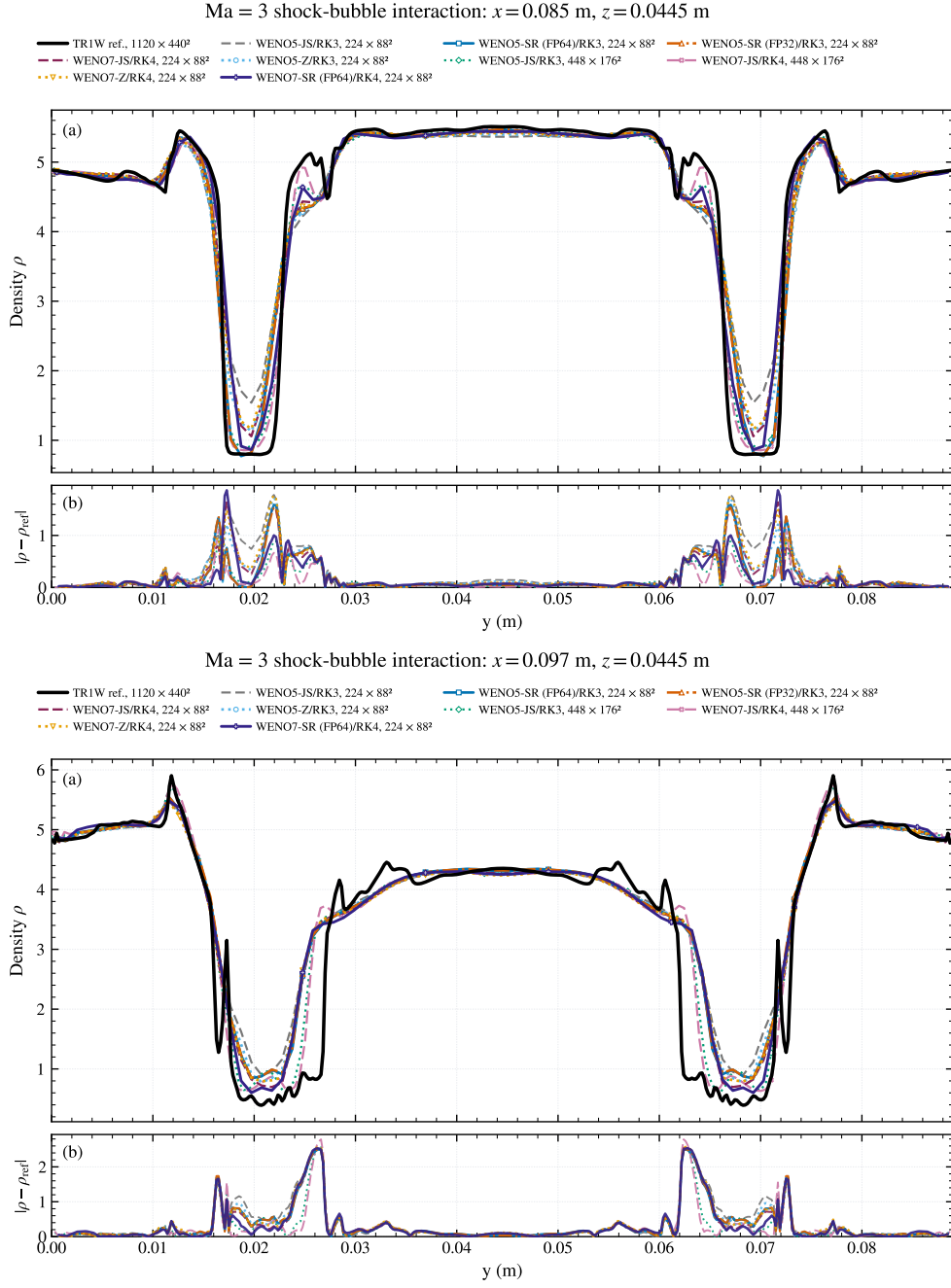


Figure 22: Central-plane density profiles and pointwise absolute errors for the three-dimensional Mach-3 shock–bubble interaction. Top: $x = 0.085$; bottom: $x = 0.097$. The black curve is the $1120 \times 440 \times 440$ WENO7–ADER4 reference. WENO5 methods use SSPRK3 and WENO7 methods use the four-stage fourth-order downwind TVD Runge–Kutta scheme.

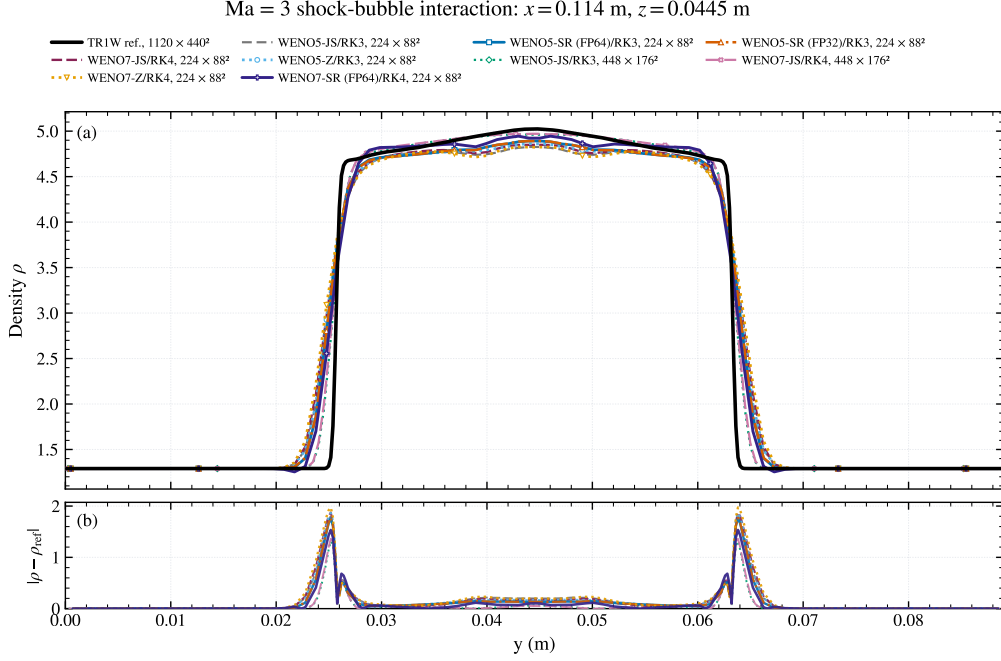


Figure 23: Central-plane density profile and pointwise absolute error at $x = 0.114$ for the three-dimensional Mach-3 shock–bubble interaction. The cut crosses the leading bubble front and highlights differences in numerical broadening. The black curve is the $1120 \times 440 \times 440$ WENO7–ADER4 reference.

The local density error reported in Table 9 is

$$E_1(x_\ell) = \int_{I_h} |\rho(y; x_\ell, z_c) - \rho_{\text{ref}}(y; x_\ell, z_c)| \, dy. \quad (52)$$

Here I_h is the common interval spanned by the candidate and reference cell centres. Candidate profiles are linearly interpolated to the reference y -coordinates, and the integral is evaluated by the trapezoidal rule.

The comparisons show three consistent trends. First, WENO5-Z and both learned WENO5 variants reduce E_1 relative to WENO5-JS on every cut. At $x = 0.085$, where the line passes through the rolled-up vortex cores, WENO5-SR FP64 and MLP-FP32 reduce the error by 36.2% and 37.5%, respectively. Their close agreement on all three cuts indicates that restricting the small MLP to FP32 does not materially change the resolved solution. Second, WENO7-Z is not uniformly more accurate than WENO7-JS in this strongly under-resolved interaction, whereas WENO7-SR reduces the WENO7-JS error by 21.9%, 3.1%, and 34.5% at the three successive cuts. It is therefore the most accurate $224 \times 88 \times 88$ method at every reported location. In particular, at the leading front ($x = 0.114$), its error is 23.0% below that of the more accurate WENO5-SR variant. Third, uniform refinement substantially lowers both classical errors, as expected; the $448 \times 176 \times 176$ results are included as resolution context rather than as like-for-like competitors to the learned $224 \times 88 \times 88$ calculations.

Table 9: Integrated local density errors E_1 for the three central-plane profiles, measured against the $1120 \times 440 \times 440$ WENO7–ADER4 reference. Bold entries identify the smallest error within each resolution group.

Cut	224 × 88 × 88							448 × 176 × 176	
	W5-JS	W5-Z	W5-SR64	W5-SR32	W7-JS	W7-Z	W7-SR64	W5-JS	W7-JS
$x = 0.085$	2.70236×10^{-2}	2.27186×10^{-2}	1.72416×10^{-2}	1.68885×10^{-2}	2.12015×10^{-2}	2.28873×10^{-2}	1.65692×10^{-2}	1.23858×10^{-2}	1.17628×10^{-2}
$x = 0.097$	3.22145×10^{-2}	2.99235×10^{-2}	2.95754×10^{-2}	3.00339×10^{-2}	2.83626×10^{-2}	3.02976×10^{-2}	2.74854×10^{-2}	2.27474×10^{-2}	2.26253×10^{-2}
$x = 0.114$	1.23214×10^{-2}	1.17924×10^{-2}	1.03177×10^{-2}	1.04390×10^{-2}	1.21228×10^{-2}	1.36931×10^{-2}	7.94580×10^{-3}	5.08478×10^{-3}	5.36308×10^{-3}

Table 10: WENO5 wall time on the 400^2 Riemann configuration C.4. Compilation, model loading, initialisation, diagnostics, and output are excluded.

Problem	Method	Mean (s)	Std. dev. (s)	Cost/JS	Overhead
C.4	WENO5-JS	6.605	0.032	1.00×	0.0%
C.4	FP64 MLP	39.333	0.280	5.96×	495.5%
C.4	FP32 MLP	14.253	0.072	2.16×	115.8%

4.10. Mixed-precision inference cost

End-to-end timings are collected for classical WENO5 and the FP64- and FP32-MLP WENO5 implementations on the 400^2 C.4 problem. Warp compilation, model loading, initialisation, diagnostics, plotting, and file output are excluded. Each configuration is warmed up and then repeated three times. Timing results are reported only if every repeat passes the same numerical validity checks.

The classical, FP64-MLP, and FP32-MLP paths require 6.605 ± 0.032 s, 39.333 ± 0.280 s, and 14.253 ± 0.072 s, respectively. Evaluating the reflection-equivariant MLP in single precision therefore accelerates the complete learned calculation by 2.76×

 and reduces its cost relative to classical WENO5 from 5.96×

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